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To Share or Not? Hybrid e-Commerce Platform's Provision and Sharing Strategies of Consumer Credit Service

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ABSTRACT

As a new financial innovation, consumer credit services are increasingly embraced by e-commerce platforms. This paper presents a stylized model to determine when an e-commerce platform should launch credit services and whether to share them with third-party sellers. Our findings indicate that the platform benefits from launching credit services only when consumer preference outweighs the delinquency rate. Furthermore, service sharing can create a win-win scenario with moderate fees and high consumer preference. Contrary to common belief, sharing services with sellers can reduce consumer surplus, suggesting that widespread use of consumer credit services is not necessarily beneficial for consumers.

1 | Introduction

In recent years, the rapid advancement of mobile technology, coupled with the surge in e-commerce, has led to the emergence of e-commerce consumer credit services. These services, such as 30-day interest-free service or equal installment interest-free service, alleviate consumers' payment burdens and facilitate seamless payment processes for online purchases, enhancing consumer satisfaction and stimulating demand growth[1] (Jia et al. 2018). Due to these advantages, numerous e-commerce giants have launched their own consumer credit services. For instance, JD.com pioneered the "JD Baitiao" consumer credit service in 2014, followed by Alibaba's introduction of "Ant Credit Pay" in January 2015 (Duan et al. 2024). Once introduced, e-commerce consumer credit services have been embraced exponentially. In China, Buy Now Pay Later (BNPL) payments are projected to grow by 14.6% in 2024, reaching \$108.89 billion. International BNPL companies like Afterpay and Klarna are also entering the Chinese market due to its immense potential (see note 1). In the United States, Amazon launched its credit service "Amazon Pay Later" in September 2018, which saw explosive growth during the COVID-19 pandemic.[2,3] A survey conducted in 2024 revealed that 49% of American consumers have utilized BNPL services, with 40% using them at least once a month.[4]

However, with the accelerated development of e-commerce consumer credit services, various risks have gradually come to light, including late repayments and bad debt losses. According to the prospectus of Ant Group, the delinquency rates for consumer credit facilitated through the Alipay platform have been steadily increasing from 2017 to 2019.[5] Additionally, a survey conducted by the Federal Reserve in 2024 revealed that 18% of Americans who have utilized credit services reported making late payments (see note 1). These issues not only raise social concerns but also pose significant challenges to the development of the platforms themselves. In practice, we observe that some platforms, such as eBay.com, have refrained from pursuing the introduction of their own consumer credit services. This observation raises an important question:Under what conditions would it be advantageous for an e-commerce platform to launch consumer credit services?

With the accelerating development of e-commerce, the operational modes of e-commerce platforms have undergone significant transformations. Many platforms, such as Amazon.com, Taobao.com, and JD.com, have evolved from a traditional single mode (reselling or agency mode) to a hybrid mode. Under the hybrid mode, the platform not only competes with third-party sellers as a retailer but also provides various operational services while charging commission fees to these sellers (Kwark, Chen, and Raghunathan 2017). Between 2022 and 2023, the revenue Amazon.com earned from third-party seller services increased by 19%.[6] Till the second quarter of 2024, third-party sellers have accounted for 61% of the products sold on Amazon.com.[7] This creates an interesting coopetition relationship between third-party sellers and the platform. Consequently, the decision to share consumer credit services with third-party sellers becomes crucial for the platform. In practice, we observe that "JD Baitiao" initially served only its self-operated (reselling) channel upon its launch, and it was not until later in 2015 that JD.com began to extend this service to third-party sellers. Additionally, we find that on Amazon.com, some sellers, such as Lenovo, are willing to utilize the consumer credit service provided by the platform, while others, such as Michael Kors, are reluctant to do so. These observations prompt us to inquire: Under what conditions is it optimal for the platform to establish a consumer credit service sharing agreement with third-party sellers? Furthermore, is consumer credit service always beneficial to consumers? How does it impact consumer surplus?

To address the aforementioned research questions, we examine a hybrid e-commerce market that includes an online platform, a wholesale supplier, a third-party seller, and a continuum of consumers. Specifically, the platform resells products purchased from the wholesale supplier, while the third-party seller enters the platform to sell a substitute product directly to consumers. We consider three alternative modes based on whether the platform launches the consumer credit service and shares it with the third-party seller: (1) no consumer credit service mode (*NN*), in which the platform does not provide any credit service; (2) launching but not sharing mode (*IN*), where the platform offers the credit service solely for its own products and does not extend this service to third-party sellers; and (3) launching and sharing mode (*II*), where the platform provides the credit service for its own products and shares it with third-party sellers. This paper advances theoretical investigations into the provision and sharing strategies of consumer credit services by hybrid e-commerce platforms, while also examining the impacts of these services on the platform itself, third-party sellers, and consumers. We will now highlight the key findings of our study.

First, we find that the willingness of the hybrid platform to launch consumer credit services critically depends on consumers' preferences for the service and the repayment rate. This is due to the trade-off between the benefits derived from the service advantages and the associated costs (risks) of realizing such benefits. The platform is motivated to launch the credit service only when consumers' preference exceeds the delinquency rate. This finding offers a potential explanation for why some hybrid platforms actively embrace consumer credit services while others do not. One contributing factor may be the platform's capacity to manage the risk of consumer defaults; the stronger this capacity, the more inclined the platform will be to introduce consumer credit services.

Next, we turn our attention to the platform's decision regarding the sharing of credit services with third-party sellers. Our analysis reveals that strategic decisions in this context are significantly influenced by consumers' preferences and the service fees that the platform charges for credit services. Consistent with expectations, if the service fee is set too low, the platform is hesitant to share the service; conversely, when the service fee is high, the platform is always inclined to share. Interestingly, when the service fee falls within an intermediate range, we find that the platform prefers to share the service only when consumer preference is high, despite the potential for increased competition. This complexity arises from the multifaceted impact of sharing credit services on the platform's operations. On one hand, sharing the service with sellers may diminish the platform's competitive advantage; on the other hand, it allows the platform to benefit from charging a service fee. Our analysis indicates that when consumer preference is elevated, the positive effects of sharing outweigh the drawbacks, leading the platform to be more willing to share its credit services.

Upon investigating the impact of credit services on consumers, we find, somewhat counterintuitively, that credit services do not always benefit consumers. When the hybrid platform introduces consumer credit services, consumer surplus is invariably greater than in scenarios without such services. This is attributable to the fact that, in comparison to the mode without consumer credit service (*NN*), the third-party seller is compelled to lower the retail price under the launching but not sharing mode (*IN*) to mitigate service disadvantages, thereby benefiting consumers. However, when the platform shares the consumer credit service with third-party sellers, consumer surplus diminishes. In this case, sharing credit services proves detrimental to consumers. The underlying reason for this finding is that when both channels utilize credit services, the retail prices of products in both channels tend to increase, ultimately leading to a reduction in overall consumer surplus.

We also extend our basic model by accounting for consumers' choices regarding credit services. Our findings remain robust with this model extension. By considering the presence of consumers who refuse to utilize credit services, our analysis indicates that the platform's decision is also influenced by the proportion of these consumers. The platform can derive benefits from launching the service only when a significant proportion of consumers are willing to engage with it. Furthermore, we observe that as this proportion increases, there is less opportunity for third-party sellers and the platform to reach a service-sharing agreement.

The rest of this study is arranged as follows. Section 2 reviews the related research and highlights the contribution. Next, we describe three different modes in Section 3. Section 4 analyzes the equilibrium results of the three modes and compares the equilibrium solutions under different modes. Section 5 extends our base model by relaxing the assumptions about consumers. Section 6 summarizes the main findings and puts forward the possible directions for future research. The proofs can be seen in the Appendix A.

2 | Literature Review

This study is closely related to the thriving literature on the interface of operations and finance. In recent years, research on this stream mainly focuses on supply chain finance, examining the impact of financial innovations—such as trade credit—on supply chain performance (e.g., Kouvelis and Zhao 2018; Lee, Zhou, and Wang 2018; Li, An, and Song 2018; Tong, DeCroix, and Song 2020; Zhang, Wu, and Liang 2018; Zhen et al. 2020). Regarding consumer credit/installment, prior literature mainly focuses on bank credit cards and their impacts on consumers' consumption behavior (e.g., Gafeeva, Hoelzl, and Roschk 2018; Gross and Souleles 2002; Runnemark, Hedman, and Xiao 2015; Soman and Cheema 2002). Prelec and Simester (2001) show that bank credits can help consumers purchase products more accessibly, thus increasing their willingness to pay. Recently, with the accelerated development of information technologies, financial scenarios have gradually evolved from offline to online environments. Some scholars have empirically studied the impacts of mobile payment on consumers' consumption behavior (e.g., Park et al. 2019; Talwar et al. 2020). However, intersectional studies on consumer credit and operations are relatively rare, particularly in relation to the online credit services offered by e-commerce platforms. The research topics and contributions of these studies are as follows. Xie et al. (2021) investigate the impact of the service on a platform supply chain, which includes a third-party seller and a platform. Niu, Zeng, and Liu (2021) consider the internet-based installment service under a chain-to-chain competition problem and investigate the motivations of different types of platforms (agent and reselling) to provide the service under competition. Zha et al. (2022) analyze the impact of consumer credit services on consumers' strategic behavior, asserting that the provision of credit services can alleviate waiting behavior among low-DPI consumers. Li, Zha, and Dong (2023) further delve into the platform's credit entry strategy in the face of credit card competition. Wei et al. (2023) introduce a concept of dual credit services, encompassing both buyer's credit service and seller's credit service, and posit that dual credit consistently confers benefits to consumers as compared to singular buyers' credit, with the effectiveness of dual credit services being contingent upon product types. Most recently, Duan et al. (2024) examine the adoption strategies for e-commerce consumer credit services in the context of platform competition, finding that channel forms and features of consumer credit services significantly influence their adoption. In contrast to these studies, our research focuses on a hybrid platform that operates both as a retailer and as a service provider for third-party sellers. We investigate the platform's motivations to introduce consumer credit services and to share this service with sellers, highlighting the platform's dual role and the strategic implications of such sharing.

Our paper further adds to the stream of literature on hybrid e-commerce platforms. In practice, e-commerce platforms typically adopt the reselling mode (e.g., BestBuy.com), agency selling mode (e.g., Zappos.com), or hybrid mode (e.g., Taobao.com, JD.com, Amazon.com). Previous studies mainly focus on the pricing and channel structure choice within pure modes, that is, reselling and agency selling mode (e.g., Chen, Duan, and Li 2022; Hagiu and Wright 2015; Hao and Fan 2014; Shen, Willems, and Dai 2019; Tan, Carrillo, and Cheng 2016; Tan and Carrillo 2017; Zhang, Nie, and Luo 2024). A recent stream of studies has increasingly concentrated on hybrid modes. For instance, Tian et al. (2018) examine the competition among upstream suppliers and identify key factors influencing online platforms' choice of operational modes. They conclude that when the cost of order fulfillment and the intensity of competition between suppliers are both moderate, the hybrid mode will be optimal for the platform. Zennyo (2020) builds upon Tian et al. (2018) by further considering the asymmetry of potential demands among upstream suppliers. They find that when the degree of product difference between suppliers is at a medium level, the hybrid mode, where only the niche supplier chooses the agency contract, may occur in equilibrium. In this work, we also consider competitive upstream suppliers who choose different contracts with the platform. Different from the aforementioned works, this work investigates the consumer credit service operated by the platform, which provides a new research perspective for hybrid e-commerce platforms.

Our work also contributes to the literature on the coopetition of the supply chain. The concept of coopetition represents the interdependent relationship between two or more companies when they simultaneously compete and cooperate (Gurnani, Erkoc, and Luo 2007). Earlier literature mainly focuses on traditional supply chains, examining the coopetition among competing manufacturers as well as between manufacturers and retailers (e.g., Bakshi and Kleindorfer 2009; Chen, Wang, and Xia 2019; Gu, Wei, and Wei 2021; Hafezalkotob 2017; Luo, Chen, and Wang 2016; Niu, Wang, and Guo 2015; Zhang and Frazier 2011). Recent works have broadened to investigate coopetition on e-commerce platforms (e.g., Feng et al. 2020; Lei, Li, and Cheng 2023; Niu et al. 2020; Qin, Liu, and Tian 2020; Wang et al. 2023; Zhang et al. 2021). For instance, Qin, Liu, and Tian (2020) investigate the logistics service-sharing strategy of hybrid platforms, where the platform engages in competition with third-party sellers while also providing service to sellers. They find that the decision to share logistics services is influenced by the logistics service level and market potential. Specifically, when both market potential and service levels are in a moderate range, a win-win situation can be achieved. Feng et al. (2020) consider a luxury fashion rental setting. They reveal the rationale behind the designer brand firms' active cooperation with their competing rental platforms. Zhang et al. (2021) investigate suppliers' encroachment strategies on e-commerce platforms and conclude that investing in platform services can deter supplier encroachment while simultaneously increasing suppliers' dependence on the platform under certain conditions. Wang et al. (2023) investigate referral strategies in a market comprising two suppliers and an e-commerce platform, and find that the recommendation market's initial size has an essential influence on the players' strategies. In contrast to these studies, our research investigate coopetition in the context of consumer credit services. Specifically, we examine how the provision and sharing of credit service of a hybrid platform impact the coopetition between the platform and the third-party seller, thereby offering valuable insights into the literature on supply chain coopetition.

Table 1 summarizes the existing studies to show the contribution of our research.

TABLE 1 | Comparison of this research with previous studies.

	Consumer credit	Service sharing	Hybrid e-commerce platform	Coopetition
Xie et al. (2021)	√			
Niu, Zeng, and Liu (2021)	√			
Zha et al. (2022)	√			
Li, Zha, and Dong (2023)	√			
Wei et al. (2023)	√			
Duan et al. (2024)	√			√
Tian et al. (2018)			√	√
Zennyo (2020)			√	√
Qin, Liu, and Tian (2020)		√		√
Feng et al. (2020)				√
Zhang et al. (2021)			√	√
Wang et al. (2023)			√	√
Niu et al. (2020)		√		√
Lei, Li, and Cheng (2023)		√		√
Our paper	√	√	√	√

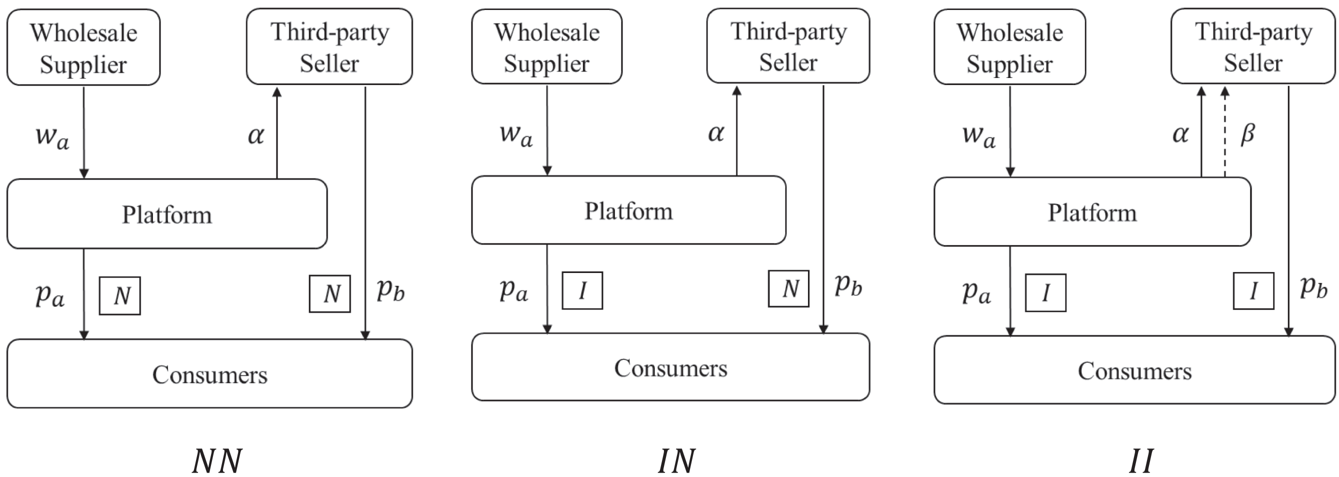

FIGURE 1 | Details of different modes.

3 | The Model

We consider an e-commerce market that includes a hybrid platform (he), a wholesale supplier, a third-party seller (she), and a continuum of consumers, in which the platform not only resells product A from the wholesale supplier, but also allows the third-party seller to sell a substitute product B directly to consumers on it. The third-party seller is required to pay a certain percentage α of her revenue to the platform, which ranges from 1% to 15% in industrial practice (Qin, Liu, and Tian 2020; Zennyo 2020). Therefore, following Tan, Carrillo, and Cheng (2016) and Geng, Tan, and Wei (2018), we assume that α is exogenous, with $0 < \alpha < 1/5$.

The consumer credit service launched by the platform can improve the convenience of consumption but may carry the risk of consumers' default. We denote the repayment rate with the parameter θ. Anecdotal evidence indicates that the timely repayment rate of credit cards is about 95.9% (Jia et al. 2016), while that for P2P (i.e., peer-to-peer lending) platforms is around 63.64% (Du et al. 2020). Consequently, we assume $1/2 < \theta < 1$. If the platform shares the consumer credit service with the third-party seller, the seller is required to pay an additional percentage β of their revenue to the platform. In practice, this percentage ranges from 1% to 5%, with Amazon charging 3.4% and Taobao 1%. Therefore, we assume $0 < \beta < 1/10$. As illustrated in Figure 1, we consider three alternative modes based on whether the platform launches and shares the consumer credit service with the third-party seller. The sequence of events for each operational mode remains the same. The wholesale supplier begins by offering a wholesale price w_a to the platform. Subsequently, the platform and the third-party seller simultaneously decide retail prices p_a and p_b for consumers.

1. No consumer credit service mode (*NN*). Under this situation, the platform does not introduce the consumer credit service. Consumers are heterogeneous in their valuation of the product v, which is uniformly distributed between 0 and 1. Note that JD.com and Amazon.com typically have advantages in self-operated product services like display, logistics, assistance, and after-sales support (Qin, Liu, and Tian 2020), which can increase consumers' willingness to pay. Therefore, we assume the consumers' valuation of products B is λv, where λ $(0 < \lambda < 1)$ represents product differentiation. Consequently, consumers' utility from purchasing products A and B can be expressed as $u_a = v - p_a$ and $u_b = \lambda v - p_b$, respectively.
2. Launching consumer credit service but no sharing mode (*IN*). In this setting, the platform only provides the consumer credit service for his self-operated product and does not share this service with the third-party seller. We use γp_a to represent the additional utility that consumers gain from the credit service, where γ represents consumers' preference for the service. The higher the preference, the greater the utility gained from the service. Furthermore, considering that consumer credit services alleviate the pressure of consumers' cash flow, we assume that the additional utility is also correlated with the product's price. The higher the price, the higher the benefits consumers receive from the consumer credit service. Hence, consumers' utility from purchasing product A is $u_a = v - p_a + \gamma p_a$, and the utility from purchasing product B is the same as *NN* mode.
3. Launching consumer credit service and sharing mode (*II*). In this setting, the platform not only provides the consumer credit service on his self-operated product, but also shares it with the third-party seller. Consumers' utility from purchasing product B is $u_b = \lambda v - p_b + \gamma p_b$, and the utility from purchasing product A is the same as *IN* mode.

We summarize all notations in Table 2. The subscripts denote different parties: the wholesale supplier (denoted by W), and the third-party seller (denoted by T), and the e-commerce platform (denoted by P). The superscripts indicate different modes, including *NN*, *IN*, and *II*.

4 | Equilibrium Analysis

According to the use of credit service by the platform and the third-party seller, we identify three modes: *NN*, *IN*, and *II*. We first consider the benchmark scenario in which neither the hybrid platform nor the third-party seller uses the consumer credit service (*NN* mode). Next, we examine the *IN* mode, where the hybrid platform exclusively provides consumer credit services for the products it resells and does not share this service with the third-party seller. By comparing the results under the *NN* and *IN* modes, we explore the conditions under which the hybrid platform is inclined to launch the consumer credit service and assess its impact on the wholesale supplier, the third-party seller, and consumers. To emphasize the critical parameter that characterizes the consumer credit service, we also investigate how the repayment rate influences equilibrium outcomes.

Finally, we address the scenario in which the platform opts to share the consumer credit service with the third-party seller, who is also interested in utilizing this service (*II* mode). By comparing results from *IN* and *II* modes, we analyze the conditions necessary for reaching an agreement on sharing credit services between the third-party seller and the hybrid platform, while further discussing how the repayment rate affects equilibrium outcomes.

4.1 | No Consumer Credit Service (*NN*)

Under this setting, consumers will purchase product A when $u_a \geq \max\{u_b, 0\}$, and they will opt for product B when $u_b \geq \max\{u_a, 0\}$. To focus on the scenario where both product A and product B exist in the market, we assume that $v_b < v_0 < 1$, where v_0 represents the indifferent point at which consumers are equally inclined to purchase either product A or product B, and v_b represents the indifferent point of whether to purchase the product B or not buying. Correspondingly, the demand of the product A is given by $D_a^{NN} = Pr\left(v_0 \leq v \leq 1\right) = 1 - \frac{p_a^{NN} - p_b^{NN}}{1-\lambda}$, and the demand of

TABLE 2 | Notations.

Notation	Description
v	Consumers' valuation of the product A
λ	Product differentiation
D_a^k	The demand of product A $(k = NN, IN, II)$
D_b^k	The demand of product B $(k = NN, IN, II)$
p_a^k	Retail price of product A, decision variable $(k = NN, IN, II)$
p_b^k	Retail price of product B, decision variable $(k = NN, IN, II)$
w_a^k	Wholesale price of product A, decision variable $(k = NN, IN, II)$
α	Proportion of the revenue that the platform charges as a commission fee
β	Fees that the platform charges to the seller for the consumer credit service
γ	Consumers' preference for the consumer credit service
θ	The repayment rate after using the consumer credit service
CS^k	Consumer surplus under different modes $(k = NN, IN, II)$
π_i^k	Profit of different parties $(i = W, T, P)$ under different modes $(k = NN, IN, II)$

the product B is $D_b^{NN} = Pr\left(v_b \le v \le v_0\right) = \frac{p_a^{NN} - p_b^{NN}}{1-\lambda} - \frac{p_b^{NN}}{\lambda}$. The profits for the wholesale supplier, the third-party seller, and the platform are as follows:

$$\pi_W^{NN} = w_a^{NN} D_a^{NN} \tag{1}$$

$$\pi_T^{NN} = (1-\alpha) p_b^{NN} D_b^{NN} \tag{2}$$

$$\pi_P^{NN} = \left(p_a^{NN} - w_a^{NN}\right) D_a^{NN} + \alpha p_b^{NN} D_b^{NN} \tag{3}$$

We use backward induction to solve this game. For a given wholesale price w_a^{NN}, we first characterize the retail prices by maximizing π_P^{NN} and π_T^{NN}. Then, by substituting these retail prices into π_W^{NN}, we determine the wholesale price w_a^{NN}. The following lemma characterizes the equilibrium results.

Lemma 1. Under the *NN* mode, the equilibrium outcomes are as follows:

i. Wholesale price: $w_a^{NN*} = \frac{(1-\lambda)(2-\alpha\lambda)}{2(2-\lambda)}$.
ii. Retail prices: $p_a^{NN*} = \frac{(1-\lambda)(6-(2+\alpha)\lambda)}{(2-\lambda)(4-\lambda-\alpha\lambda)}$; $p_b^{NN*} = \frac{(1-\lambda)\lambda(6-(2+\alpha)\lambda)}{2(2-\lambda)(4-\lambda-\alpha\lambda)}$.
iii. Demand: $D_a^{NN*} = \frac{2-\alpha\lambda}{2(4-\lambda-\alpha\lambda)}$; $D_b^{NN*} = \frac{1}{4-2\lambda} + \frac{1}{8-2(1+\alpha)\lambda}$.
iv. Profits: $\pi_W^{NN*} = \frac{(1-\lambda)(2-\alpha\lambda)^2}{4(2-\lambda)(4-\lambda-\alpha\lambda)}$; $\pi_T^{NN*} = \frac{(1-\alpha)(1-\lambda)\lambda(6-(2+\alpha)\lambda)^2}{4(2-\lambda)^2(4-\lambda-\alpha\lambda)^2}$; $\pi_P^{NN*} = \frac{(1-\lambda)(16-4(4-\lambda)\lambda+\alpha\lambda(44-4(7-\lambda)\lambda-\alpha\lambda(24+\lambda(12-\alpha(3-\lambda)+\lambda))))}{4(2-\lambda)^2(4-\lambda-\alpha\lambda)^2}$.
v. Consumer surplus:

$$CS^{NN*} = \frac{16+\lambda\left(68-12(5-\lambda)\lambda+\alpha^2\lambda\left(4+\lambda-\lambda^2\right)-4\alpha(4+\lambda(7-(6-\lambda)\lambda))\right)}{8(2-\lambda)^2(4-\lambda-\alpha\lambda)^2}.$$

Intuitively, we find that with the increase of commission fee α, the profit of the third-party seller will decrease while the platform's profit will increase. An intriguing finding from Lemma 1 is that the wholesale supplier's profit also diminishes with α (i.e., $\frac{\partial \pi_W^{NN*}}{\partial \alpha} < 0$). The intuition behind this *negative spillover effect* is that as α increases, the platform can receive higher commission revenue from the third-party seller, who then responds by charging a higher retail price for product B (i.e., $\frac{\partial p_b^{NN*}}{\partial \alpha} > 0$). In turn, the platform sets a higher retail price for product A, resulting in fewer consumers purchasing product A (i.e., $\frac{\partial p_a^{NN*}}{\partial \alpha} > 0$, $\frac{\partial D_a^{NN*}}{\partial \alpha} < 0$). Ultimately, this leads to a reduction in the wholesale supplier's profit. This spillover effect is a characteristic feature of hybrid e-commerce platforms. Seminal research on hybrid platforms by Tian et al. (2018) also finds a similar spillover effect.

4.2 | Launching Consumer Credit Service but No Sharing Mode (*IN*)

In this setting, only the hybrid platform provides the consumer credit service for the products he resells, and the seller does not use the credit service. Specifically, we use γp_a^{IN} to represent the utility that consumers can receive from the consumer credit service. Correspondingly, the demand of product A is $D_a^{IN} = 1 - \frac{(1-\gamma)p_a^{IN} - p_b^{IN}}{1-\lambda}$, and the demand of product B is $D_b^{IN} = \frac{(1-\gamma)p_a^{IN} - p_b^{IN}}{1-\lambda} - \frac{p_b^{IN}}{\lambda}$. The profits of the wholesale supplier, the third-party seller and the platform are as follows, where θ represents the repayment rate of consumers who use the consumer credit service.

$$\pi_W^{IN} = w_a^{IN} D_a^{IN} \tag{4}$$

$$\pi_T^{IN} = (1-\alpha) p_b^{IN} D_b^{IN} \tag{5}$$

$$\pi_P^{IN} = \left(\theta p_a^{IN} - w_a^{IN}\right) D_a^{IN} + \alpha p_b^{IN} D_b^{IN} \tag{6}$$

Similar to *NN* mode, we use backward induction to solve this game. The results of *IN* mode are presented in the following lemma.

Lemma 2. When the platform only provides the consumer credit service for the products he resells, the equilibrium outcomes are as follows:

i. Wholesale price: $w_a^{IN*} = \frac{(1-\lambda)(2\theta-\alpha(1-\gamma)\lambda)}{2(1-\gamma)(2-\lambda)}$.
ii. Retail prices: $p_a^{IN*} = \frac{(1-\lambda)(2\theta(3-\lambda)-\alpha(1-\gamma)\lambda)}{(1-\gamma)(2-\lambda)(\theta(4-\lambda)-\alpha(1-\gamma)\lambda)}$; $p_b^{IN*} = \frac{(1-\lambda)\lambda(2\theta(3-\lambda)-\alpha(1-\gamma)\lambda)}{2(2-\lambda)(\theta(4-\lambda)-\alpha(1-\gamma)\lambda)}$.
iii. Demand: $D_a^{IN*} = \frac{2\theta-\alpha(1-\gamma)\lambda}{2(\theta(4-\lambda)-\alpha(1-\gamma)\lambda)}$; $D_b^{IN*} = \frac{1}{4-2\lambda} + \frac{\theta}{2(\theta(4-\lambda)-\alpha(1-\gamma)\lambda)}$.
iv. Profits: $\pi_W^{IN*} = \frac{(1-\lambda)(2\theta-\alpha(1-\gamma)\lambda)^2}{4(1-\gamma)(2-\lambda)(\theta(4-\lambda)-\alpha(1-\gamma)\lambda)}$; $\pi_T^{IN*} = \frac{(1-\alpha)(1-\lambda)\lambda(2\theta(3-\lambda)-\alpha(1-\gamma)\lambda)^2}{4(2-\lambda)^2(\theta(4-\lambda)-\alpha(1-\gamma)\lambda)^2}$; $\pi_P^{IN*} = \frac{(1-\lambda)\left(4\theta^3(2-\lambda)^2+\alpha^3(1-\gamma)^3(3-\lambda)\lambda^3-\alpha^2(1-\gamma)^2\theta\lambda^2(24-(12-\lambda)\lambda)+4\alpha(1-\gamma)\theta^2\lambda(11-(7-\lambda)\lambda)\right)}{4(1-\gamma)(2-\lambda)^2(\theta(4-\lambda)-\alpha(1-\gamma)\lambda)^2}$.
v. Consumer surplus:

$$CS^{IN*} = \frac{\alpha^2(1-\gamma)^2\lambda^2(4+(1-\lambda)\lambda)-4\alpha(1-\gamma)\theta\lambda(4+\lambda(7-(6-\lambda)\lambda))+4\theta^2(4+\lambda(17-3(5-\lambda)\lambda))}{8(2-\lambda)^2(\theta(4-\lambda)-\alpha(1-\gamma)\lambda)^2}.$$

Similar to Lemma 1, we find that under *IN* mode, the negative spillover effect still exists (i.e., $\frac{\partial \pi_W^{IN*}}{\partial \alpha} < 0$). Next, to focus on the critical parameter that describes the consumer credit service, we analyze the impact of the repayment rate on equilibrium outcomes.

Lemma 3. As the repayment rate θ increases, we have:

i. The retail prices p_a^{IN*} and p_b^{IN*} decrease, while the wholesale price w_a^{IN*} increases.
ii. Product A's demand D_a^{IN*} increases, while product B's demand D_b^{IN*} decreases.
iii. The platform's profit π_P^{IN*} and the wholesale supplier's profit π_W^{IN*} increase, while the third-party seller's profit π_T^{IN*} decreases.
iv. Consumer surplus CS^{IN*} increases.

As the repayment rate θ increases, both retail prices decrease. This occurs because a higher repayment rate reduces the platform's exposure to the risk of consumer default, thereby incentivizing the platform to set a lower retail price to boost demand of product A. In response to the competitive pressure from product A, the third-party seller is compelled to lower the price of product B. We refer to this phenomenon, where retail prices decline in response to an increased repayment rate, as the *strategic effect of asymmetric consumer credit services*. Interestingly, we find that the demand of product B continues to be cannibalized even when its price is reduced. Additionally, as the repayment rate θ increases, the wholesale supplier raises the wholesale price despite the platform lowering the retail price. The rationale behind this finding is that when the repayment rate is high, the wholesale supplier anticipates that the platform will seek to enhance demand of product A. To capitalize on the benefits of the platform's consumer credit service, the supplier opts to increase the wholesale price.

Lemma 3 shows that as the repayment rate increases, the platform strategically reduces the price of product A. This price reduction compels the third-party seller to also decrease the price of product B, effectively diminishing the third-party seller's profit. Although the supplier is not directly involved in the consumer credit service, they can indirectly benefit from the elevated repayment rate. Moreover, consumers stand to gain from the reduced retail prices resulting from the strategic implications of the asymmetric consumer credit service. This finding yields meaningful implications for platforms that have implemented consumer credit services. Strengthening relationships with consumers to enhance loyalty and reduce default risks is crucial. Additionally, since consumers can also benefit from a higher repayment rate, we recommend that platforms provide financial literacy programs to help consumers understand the importance of timely repayments. By implementing these strategies, platforms can create a more efficient and beneficial credit service environment.

Based on Lemmas 1 and 2, we next compare the platform's profits under the two modes (i.e., *NN* and *IN*) and investigate the condition that the platform is willing to provide consumer credit service, as detailed in the following proposition.

Proposition 1. By comparing the platform's profit under the *NN* and *IN* modes, we have:

i. When $\gamma > 1 - \theta$, the platform will launch the consumer credit service;
ii. When $\gamma \leq 1 - \theta$, the platform will not launch the consumer credit service.

Proposition 1 highlights that the platform's decision to launch a consumer credit service critically depends on consumers' preferences for the service and the repayment rate. By introducing the credit service, the platform can enhance its product advantages, thereby increasing demand of the products it resells. However, it must also bear the risk of consumer defaults. Thus, the decision to launch the service reflects a trade-off between the benefits of service advantage and the associated risks. In practice, some hybrid online platforms (e.g., Amazon.com and JD.com) have introduced their own consumer credit services, while others (e.g., eBay.com) have not. Our findings suggest that a key factor in this decision may be the platform's ability to manage consumer default risks. The stronger this ability, the more inclined the platform is to offer consumer credit services. This insight provides important managerial implications for platform operations. It suggests that blindly launching consumer credit services to follow trends may not always be beneficial. Therefore, having sufficient consumer information and the capability to dynamically assess consumers' repayment abilities are crucial prerequisites for successfully launching such services.

Under this condition, we can get the profit comparison of the wholesale supplier and the third-party seller in different modes. We can also analyze the impact of the platform's decision from the perspective of prices, demands, and consumer surplus, as shown in Proposition 2.

Proposition 2. Under the condition $\gamma > 1 - \theta$, by comparing the equilibrium results under *IN* and *NN* modes, we have:

i. $p_a^{IN*} > p_a^{NN*}$; $w_a^{IN*} > w_a^{NN*}$; $p_b^{IN*} < p_b^{NN*}$.
ii. $D_a^{IN*} > D_a^{NN*}$; $D_b^{IN*} < D_b^{NN*}$.
iii. $\pi_W^{IN*} > \pi_W^{NN*}$; $\pi_T^{IN*} < \pi_T^{NN*}$.
iv. $CS^{IN*} > CS^{NN*}$.

These results essentially demonstrate that the platform can raise the retail price of product A without losing demand when introducing the consumer credit service, as it enhances consumption convenience. This strengthens the platform's competitive advantage. Meanwhile, the wholesale price can be higher than in the *NN* mode to capitalize on product A's competitive edge. However, launching the consumer credit service adversely affects the third-party seller. To compensate for her product's service disadvantage, the third-party seller must lower the retail price of product B. Proposition 2 indicates that even with a reduced price, demand of product B still declines due to the impact of the consumer credit service. Furthermore, we find that due to the reduced retail price of product B, when the hybrid platform introduces the consumer credit service, consumer surplus is consistently higher than in the *NN* mode, indicating that consumers benefit from this initiative.

The introduction of consumer credit services exclusively for consumers buying product A can benefit both the wholesale supplier and consumers, while the third-party seller is disadvantaged due to increased competition. Therefore, we recommend that third-party sellers remain vigilant about platform announcements regarding credit services. They should consider offering differentiated products or services to attract consumers, such as personalized shopping experiences and efficient fulfillment processes. For the platform, it is advisable to implement programs that support third-party sellers in adapting to the new competitive environment. This could include providing additional value-added services to help maintain their competitiveness.

4.3 | Launching Consumer Credit Service and Sharing Mode (*II*)

Under this scenario, the third-party seller and the platform reach the consumer credit service sharing agreement. That is, the platform is willing to share the consumer credit service with the third-party seller and the seller also wants to use the service. Correspondingly, the demand of product A is $D_a^{II} = 1 - \frac{(1-\gamma)(p_a^{II} - p_b^{II})}{1-\lambda}$, and the demand of product B is $D_b^{II} = \frac{(1-\gamma)(p_a^{II} - p_b^{II})}{1-\lambda} - \frac{(1-\gamma)p_b^{II}}{\lambda}$. The profits for the wholesale supplier, the third-party seller, and the platform are as follows, where β represents the service fee that the third-party seller needs to pay to the platform for using the consumer credit service.

$$\pi_W^{II} = w_a^{II} D_a^{II} \tag{7}$$

$$\pi_T^{II} = p_b^{II} D_b^{II} - (\alpha + \beta) p_b^{II} D_b^{II} \tag{8}$$

$$\pi_P^{II} = \left(\theta p_a^{II} - w_a^{II}\right) D_a^{II} + (\alpha + \beta) p_b^{II} D_b^{II} - (1 - \theta) p_b^{II} D_b^{II} \tag{9}$$

By utilizing backward induction, we solve *II* mode, and the equilibrium solutions are as follows.

Lemma 4. When both channels use the consumer credit service, the equilibrium outcomes are as follows:

i. Wholesale price: $w_a^{II*} = \frac{(1-\lambda)(2\theta + (M-\theta)\lambda)}{2(1-\gamma)(2-\lambda)}$.
ii. Retail prices: $p_a^{II*} = \frac{(1-\lambda)(3\theta(2-\lambda) + M\lambda)}{(1-\gamma)(2-\lambda)(2\theta(2-\lambda) + M\lambda)}$; $p_b^{II*} = \frac{(1-\lambda)\lambda(3\theta(2-\lambda) + M\lambda)}{2(1-\gamma)(2-\lambda)(2\theta(2-\lambda) + M\lambda)}$.
iii. Demand: $D_a^{II*} = \frac{2\theta + (M-\theta)\lambda}{4\theta(2-\lambda) + 2M\lambda}$; $D_b^{II*} = \frac{1}{4-2\lambda} + \frac{\theta}{4\theta(2-\lambda) + 2M\lambda}$.
iv. Profits: $\pi_W^{II*} = \frac{(1-\lambda)(2\theta + (M-\theta)\lambda)^2}{4(1-\gamma)(2-\lambda)(2\theta(2-\lambda) + M\lambda)}$; $\pi_T^{II*} = \frac{(1-\alpha-\beta)(1-\lambda)\lambda(3\theta(2-\lambda)+M\lambda)^2}{4(1-\gamma)(2-\lambda)^2(2\theta(2-\lambda)+M\lambda)^2}$; $\pi_P^{II*} = \frac{(1-\lambda)\left(2(\theta-M)\lambda(3\theta(2-\lambda)+M\lambda)^2 - (4-2\lambda)(2\theta+(M-\theta)\lambda)\left(M^2\lambda^2 - 2\theta^2(2-\lambda)(1+\lambda) + M\theta\lambda(4-3\lambda)\right)\right)}{8(1-\gamma)(2-\lambda)^2(2\theta(2-\lambda)+M\lambda)^2}$.
v. Consumer surplus: $CS^{II*} = \frac{M^2\lambda^2(4 + (1-\lambda)\lambda) + 2M\theta(2-\lambda)\lambda(4 + \lambda(7-3\lambda)) + \theta^2(2-\lambda)^2(4 + \lambda(17-5\lambda))}{8(2-\lambda)^2(2\theta(2-\lambda)+M\lambda)^2}$,

where $M = 1 - \alpha - \beta$.

When the platform shares the consumer credit service with the third-party seller and charges a service fee, Lemma 4 shows that similar to the commission fee rate α, the service fee rate β also has a *negative spillover effect* on the wholesale supplier's profit. That is, an increase in β will lead to a decrease in the wholesale supplier's profit. The reason behind this is similar to α. Next, we focus on the impact of the repayment rate θ on equilibrium results.

Lemma 5. As the repayment rate θ increases, we have:

i. The retail prices p_a^{II*} and p_b^{II*} and the wholesale price w_a^{II*} all increase.
ii. Product *A*'s demand D_a^{II*} decreases, while product *B*'s demand D_b^{II*} increases.
iii. The platform's profit π_P^{II*}, the wholesale supplier's profit π_W^{II*} and the third-party seller's profit π_T^{II*} all increase.
iv. Consumer surplus CS^{II*} decreases.

Interestingly, under *II* mode, the impact of the repayment rate on retail prices exhibits an opposite trend compared to that observed in *IN* mode, where both retail prices decrease. Unlike the findings in Lemma 3, when both the third-party seller and platform utilize the consumer credit service, the platform is inclined to collect a service fee β from the third-party seller to boost its profits. In this scenario, as the repayment rate increases, the platform experiences enhanced marginal revenue from product *B*. Consequently, the strategy of compelling the third-party seller to reduce prices becomes less advantageous. Instead, the platform is incentivized to increase the retail price of product *A*, encouraging the seller to charge a higher price for product *B*. We refer to this dynamic as the *strategic effect of the symmetric consumer credit service*. Although both products become more expensive, the price advantage of product *B* increases (i.e., $\frac{\partial(p_a - p_b)}{\partial\theta} > 0$), potentially boosting demand of product *B* and decreasing that of product *A*. As a result, the market is better segmented, allowing both the platform and the third-party seller to benefit from an increased repayment rate. Similar to the *IN* mode, as the repayment rate θ rises, the wholesale supplier will increase the wholesale price to capitalize further on the consumer credit service sharing mode.

Unlike the *IN* mode, consumer surplus decreases with the repayment rate due to increased retail prices for both products. This indicates that under the *II* mode, where consumer credit services are available for both products, a higher repayment rate can heighten the risk of consumers being exploited by the platform and the third-party seller. Consequently, when consumer credit services are shared with third-party sellers, controlling default risk becomes even more critical. At this point, the platform should be vigilant against malicious non-repayment behavior by consumers. Robust credit risk assessment models should be developed to ensure the sustainability of the credit program and minimize potential losses.

Having analyzed the equilibrium results of *IN* mode and *II* mode, we can now investigate the condition that the platform prefers to share consumer credit service with the third-party seller and she will also be willing to use the service.

Proposition 3. Under the condition that the platform has launched the installment service (i.e., $\gamma > 1 - \theta$), by comparing the equilibrium results of profits, we have:

i. If $\beta \le 1 - \alpha - \theta$, the platform is reluctant to share the consumer credit service with the third-party seller.
ii. If $1 - \alpha - \theta < \beta < (1-\alpha)(1-\theta)$, the platform is willing to share the consumer credit service only when $\gamma > \frac{1-\theta-\beta}{\alpha}$, while the third-party seller is always willing to use the consumer credit service.
iii. If $\beta \ge (1-\alpha)(1-\theta)$, the platform is always willing to share the consumer credit service, while the third-party seller is willing to use it only when $\gamma > \gamma_0$.

Proposition 3 indicates that strategic decisions primarily depend on consumers' preferences and the service fee charged by the platform to the third-party seller. As expected, if the service fee β is too low, the platform is reluctant to share the service. Conversely, with a high β, the platform prefers to share the consumer credit service. Interestingly, our analysis reveals that when the service fee is at an intermediate level, the platform opts to share the service only when consumer preference is high (i.e., $\gamma > \frac{1-\theta-\beta}{\alpha}$). Conventional wisdom suggests that higher consumer preference implies a greater competitive advantage for the platform when enjoying exclusive access to the consumer credit service. So why would the platform choose to share it with the third-party seller? Intuitively, sharing with the third-party seller means losing some competitive edge. However, it also allows the platform to benefit from charging a service fee. Higher consumer preference enables the third-party seller to set higher retail prices under *II* mode, thereby increasing benefits for the platform. Our findings demonstrate that with higher consumer preference, the positive effects prevail, prompting the platform to share the service.

For the third-party seller, utilizing the consumer credit service provided by the platform enhances her product's competitiveness. She will always be willing to use the service if the service fee β is low. However, if the service fee is high, she will only opt for the service when consumer preference for it is relatively strong (i.e., $\gamma > \gamma_0$). It is important to note that consumer preference for credit services varies with different products. For example, consumers typically have a high preference for such services with luxury accessories and electronic products, while preference tends to be lower for daily necessities. Our findings elucidate why some third-party sellers on hybrid platforms actively adopt the consumer credit service offered by the platform, whereas others choose not to. Figure 2 illustrates the main findings of Proposition 3. It shows that when the service fee is not excessively low and consumer preference for the credit service surpasses a certain threshold, both the platform prefers to share the service and the seller is willing to use it, as depicted in region *I* of Figure 2.

Proposition 3 offers insights for platforms on providing consumer credit services. The platform should carefully tailor its consumer credit strategies based on market conditions and consumer behavior. When dealing with a medium service fee, the platform should enhance its data analytics capabilities to continuously assess consumer preferences and market trends. Additionally, fostering open communication channels with third-party sellers is crucial, as it provides valuable feedback and helps platforms refine their credit services to meet the needs of both sellers and consumers.

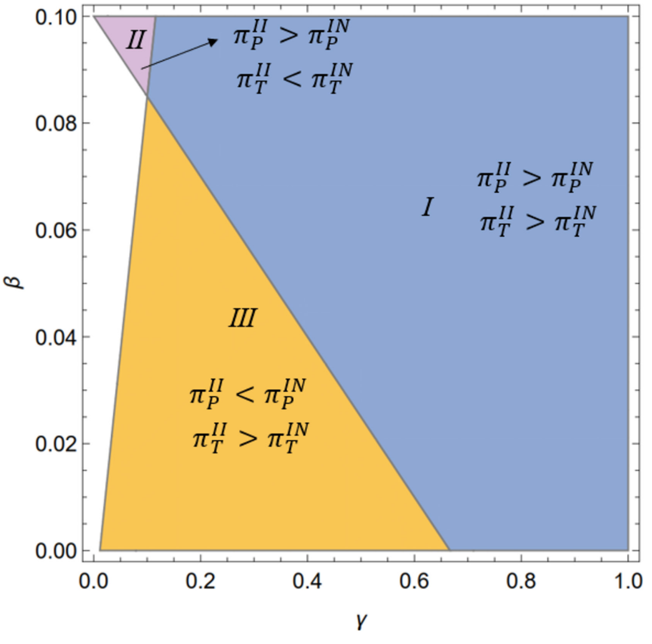

FIGURE 2 | Strategy choices for the platform and third-party seller. $\alpha = 0.15, \theta = 0.9, \lambda = 0.6$.

Under the condition that both channels use the consumer credit service, we can get the profit comparison of the wholesale supplier under the *II* and *IN* modes. We can also analyze the impact of the service-sharing agreement from the perspective of prices, demands, and consumer surplus, as shown in Proposition 4.

Proposition 4. Under the condition that the platform is willing to share the service and the third-party seller is willing to use the service, by comparing the equilibrium results under the *II* and *IN* modes, we have:

i. $p_a^{II*} > p_a^{IN*}$; $w_a^{II*} < w_a^{IN*}$; $p_b^{II*} > p_b^{IN*}$.
ii. $D_a^{II*} < D_a^{IN*}$; $D_b^{II*} > D_b^{IN*}$.
iii. $\pi_W^{II*} < \pi_W^{IN*}$.
iv. $CS^{II*} < CS^{IN*}$.

Sharing consumer credit services can alleviate price competition and benefit both the platform and the third-party seller, although it reduces the profit of the wholesale supplier and consumer surplus. For the third-party seller, using the consumer credit service significantly increases the demand of product *B* due to its service advantage, while partially cannibalizing the demand of product *A* resold by the platform. The platform has two revenue streams: profit from product *A* and service fees from the seller. Despite the cannibalization of product *A*'s market due to service sharing, the platform tends to raise product *A*'s price to potentially boost demand of product *B*, thereby increasing the service fee received from the third-party seller. It is important to note that when the platform and seller agree to share credit services, it negatively impacts the wholesale supplier. In response to decreased demand of product *A*, the wholesale supplier may lower the wholesale price, ultimately leading to a loss in profit. Therefore, we recommend that platforms develop strategies to mitigate these negative effects, such as offering additional services or collaborating on product innovation to maintain supplier engagement and ensure a stable supply chain.

Furthermore, we note that when the platform shares the credit service with the third-party seller, consumer surplus decreases due to increased retail prices. While it is commonly believed that sharing consumer credit services allows consumers to enjoy the convenience of consumption from both channels and benefit them compared to asymmetric credit services, our findings reveal that this sharing can actually be detrimental to consumers. Our analysis shows that the platform may counterintuitively raise the retail price of product *A* to boost its revenue. For high-value consumers purchasing product *A*, their consumer surplus is significantly reduced, ultimately decreasing the total consumer surplus. To mitigate this, the platform should aim to attract consumers through other means, such as enhancing service quality, offering exclusive benefits to credit service users, or developing personalized shopping experiences.

5 | Extension

In this section, we extend our base model by considering consumers' different choices regarding credit services. In the previous analysis, we assumed that consumers would always embrace the interest-free consumer credit service once offered. However, in reality, some consumers choose not to use these services. For instance, many consumers distrust the platform and worry about the negative impact on their credit history due to late payments. Consequently, despite the service's availability and convenience, these consumers are reluctant to use it. We refer to these individuals as conservative consumers.

We assume that δ proportion of consumers are the same as that in the base model, who will use the consumer credit service once it is provided, and $1-\delta$ proportion of consumers are conservative, where $0<\delta<1$. We use the superscripts "*INC*" and "*IIC*" to represent the expansion of the two basic modes "*IN*" and "*II*" respectively. For these conservative consumers, the utility of buying product A and product B are $u_a = v - p_a^{INC}$ and $u_b = \lambda v - p_b^{INC}$, respectively. In *INC* mode, for product A, the demands of consumers who use the consumer credit service and conservative consumers are $D_{a1}^{INC} = \left(1 - \frac{(1-\gamma)p_a^{INC} - p_b^{INC}}{1-\lambda}\right)\delta$ and $D_{a2}^{INC} = \left(1 - \frac{p_a^{INC} - p_b^{INC}}{1-\lambda}\right)(1-\delta)$, respectively. The demand of product B are $D_{b1}^{INC} = \left(\frac{(1-\gamma)p_a^{INC} - p_b^{INC}}{1-\lambda} - \frac{p_b^{INC}}{\lambda}\right)\delta$ and $D_{b2}^{INC} = \left(\frac{p_a^{INC} - p_b^{INC}}{1-\lambda} - \frac{p_b^{INC}}{\lambda}\right)(1-\delta)$, respectively. The profits of the wholesale supplier, the third-party seller and the platform are as follows:

$$\pi_W^{INC} = w_a^{INC}\left(D_{a1}^{INC} + D_{a2}^{INC}\right) \tag{10}$$

$$\pi_T^{INC} = (1-\alpha)p_b^{INC}\left(D_{b1}^{INC} + D_{b2}^{INC}\right) \tag{11}$$

$$\begin{aligned}\pi_P^{INC} = {} & \left(\theta p_a^{INC} - w_a^{INC}\right)D_{a1}^{INC} + \left(p_a^{INC} - w_a^{INC}\right)D_{a2}^{INC} \\ & + \alpha p_b^{INC}\left(D_{b1}^{INC} + D_{b2}^{INC}\right)\end{aligned} \tag{12}$$

The equilibrium results of the *INC* mode are detailed in the Appendix. Given the complexity of these results, which makes analysis and comparison challenging, we conducted extensive numerical studies to explore the impact of different consumer segments. A key question is how conservative consumers influence the platform's decision to launch the consumer credit service. As stated in Proposition 1, if consumers' preference for the service outweighs their delinquency rate (i.e., $\gamma > 1-\theta$), the platform will opt to launch the credit service without considering conservative consumers. However, numerical analysis reveals that the proportion of conservative consumers significantly impacts this decision. As shown in Figure 3, even when $\gamma > 1-\theta$, the platform will only be inclined to introduce the consumer credit service if the proportion of consumers who embrace it exceeds a certain threshold.

Similar to *INC* mode, in *IIC* mode, for product A, the demands of consumers who use the consumer credit service and conservative consumers can be derived as $D_{a1}^{IIC} = \left(1 - \frac{(1-\gamma)\left(p_a^{IIC} - p_b^{IIC}\right)}{1-\lambda}\right)\delta$ and $D_{a2}^{IIC} = \left(1 - \frac{p_a^{IIC} - p_b^{IIC}}{1-\lambda}\right)(1-\delta)$ respectively. For product B, the demands can be derived as $D_{b1}^{IIC} = \left(\frac{(1-\gamma)\left(p_a^{IIC} - p_b^{IIC}\right)}{1-\lambda} - \frac{(1-\gamma)p_b^{IIC}}{\lambda}\right)\delta$ and $D_{b2}^{IIC} = \left(\frac{p_a^{IIC} - p_b^{IIC}}{1-\lambda} - \frac{p_b^{IIC}}{\lambda}\right)(1-\delta)$, respectively. The profits of

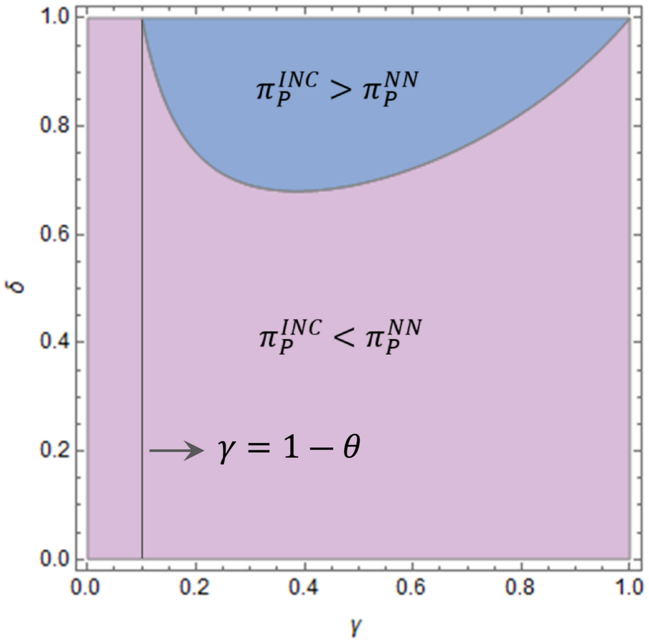

FIGURE 3 | Strategy choices for the platform when considering different consumer segments. $\alpha = 0.15, \theta = 0.9, \lambda = 0.6$.

the wholesale supplier, the third-party seller and the platform are as follows:

$$\pi_W^{IIC} = w_a^{IIC}\left(D_{a1}^{IIC} + D_{a2}^{IIC}\right) \tag{13}$$

$$\pi_T^{IIC} = (1-\alpha-\beta)p_b^{IIC}D_{b1}^{IIC} + (1-\alpha)p_b^{IIC}D_{b2}^{IIC} \tag{14}$$

$$\begin{aligned}\pi_P^{IIC} = {} & \left(\theta p_a^{IIC} - w_a^{IIC}\right)D_{a1}^{IIC} + \left(p_a^{IIC} - w_a^{IIC}\right)D_{a2}^{IIC} \\ & + (\alpha+\beta-(1-\theta))p_b^{IIC}D_{b1}^{IIC} + \alpha p_b^{IIC}D_{b2}^{IIC}\end{aligned} \tag{15}$$

The equilibrium results of the *IIC* mode are detailed in the Appendix A. We then conduct numerical experiments to explore how conservative consumers influence the platform's decision to share the consumer credit service and the third-party seller's decision to use it. Figure 4 provides several representative examples with different values of δ. Overall, we observe that considering conservative consumer behavior does not affect the main results of the base model. Whether the seller and platform reach a consumer credit service sharing agreement primarily depends on consumers' preference for the service (γ) and the service fee rate (β). For the platform, when the service fee rate β charged to the third-party seller is relatively high and consumers have a strong preference for the service, the platform prefers to share it with third parties (regions *I* and *II*). For the third-party seller, when consumer preference for the credit service is high and β is relatively low, they are more inclined to use it (regions *I* and *III*). Furthermore, we find that as the proportion of consumers who embrace the credit service (δ) increases, the space for reaching a service-sharing agreement (region *I*) becomes smaller. This is because as more consumers use the service, the platform incurs more losses than gains from sharing it, making it increasingly reluctant to do so.

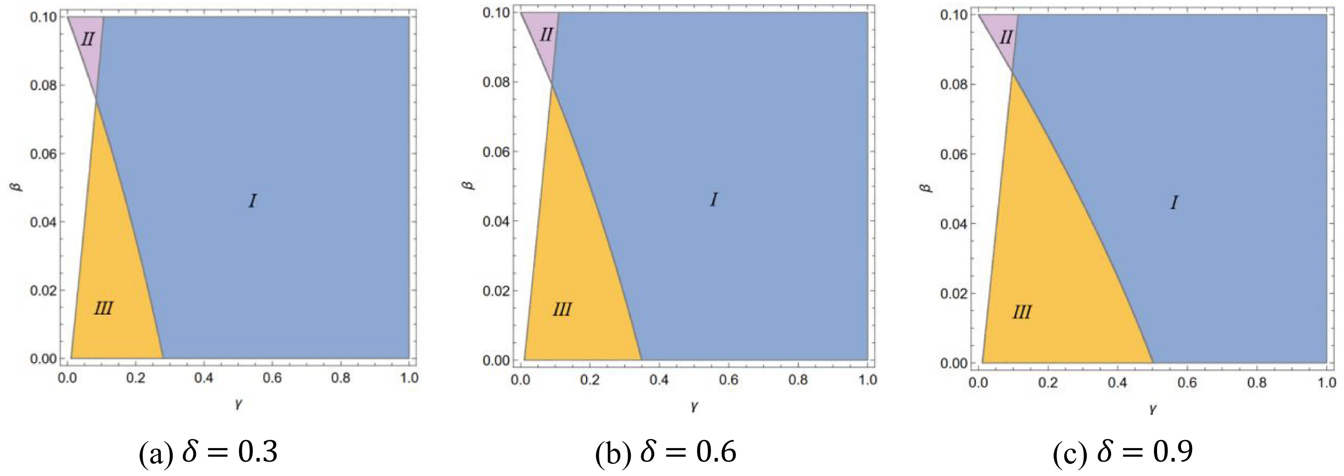

FIGURE 4 | Strategy choices under different proportions of conservative consumers. Region I represents $\pi_P^{IIC*} > \pi_P^{INC*}$ and $\pi_B^{IIC*} > \pi_B^{INC*}$; region II represents $\pi_P^{IIC*} > \pi_P^{INC*}$ and $\pi_B^{IIC*} < \pi_B^{INC*}$; region III represents $\pi_P^{IIC*} < \pi_P^{INC*}$ and $\pi_B^{IIC*} > \pi_B^{INC*}$; $\alpha = 0.15, \theta = 0.9, \lambda = 0.6$.

6 | Conclusions

With the rapid development of e-commerce in recent years, consumer credit services initiated by various e-commerce hybrid platforms have emerged as a significant trend in financial innovation. Compared to traditional bank credit services, consumer credit services offer advantages such as low thresholds and flexible limits, making them increasingly popular among consumers. These services enhance the consumer experience and boost demand of platforms. However, platforms offering consumer credit services must also bear the risk of bad debt losses. Therefore, deciding whether to provide such services is a crucial decision for e-commerce platforms. Additionally, within hybrid platforms, determining whether to share consumer credit services with third-party sellers is also vital. This paper develops a stylized model to analytically investigate the platform's strategies for providing and sharing consumer credit services. By analyzing equilibrium results under three different modes, we further examine the impact of consumer credit services on the platform, third-party sellers, and consumers. Our analysis provides several important insights into the platform's credit service operations by addressing the following key questions.

When would it be beneficial for a hybrid platform to launch consumer credit services? The platform is motivated to launch the credit service only when consumers' preference exceeds the delinquency rate. The managerial implication is that the core of operating this innovative financial service lies in the ability to control consumer default risk. For e-commerce platforms considering launching consumer credit services, having adequate consumer information and the capability to dynamically assess consumers' repayment abilities are essential requirements. This aligns with the successful experiences of many platforms. For instance, through extensive data collection and the use of artificial intelligence, Alipay maintains a low loan delinquency rate of roughly 1% to 2%, which supports Alibaba in effectively utilizing consumer credit services on Taobao.com and Tmall.com.[8]

·When is it optimal for both the third-party seller and the platform to reach the consumer credit service sharing agreement? Our findings indicate that this strategic decision critically depends on two parameters: consumers' preference for the services and the service fee. If the service fee is relatively low, the platform is unlikely to share the service. Conversely, if the platform charges a high service fee and consumer preference is low, the seller will not opt to use the service. However, when the fee is moderate and consumer preference is relatively high, reaching a consumer credit service sharing agreement becomes beneficial, achieving a win-win situation. Although sharing the service allows the platform to earn from service fees, it reduces its market advantage. With higher consumer preference for the credit service, the financial gains from service fees outweigh the loss of market advantage, making the platform more likely to share the service with sellers. This result explains why some third-party sellers on hybrid platforms choose to use consumer credit services while others do not. We also recommend that platforms enhance their data analytics capabilities to continuously assess consumer preferences and market trends, allowing them to adjust their strategies for offering and sharing consumer credit services effectively.

·Is access to consumer credit services always beneficial to consumers? When a hybrid platform launches consumer credit services exclusively for its self-operated products, consumer surplus increases compared to having no credit service. However, if both channels utilize the service, contrary to intuition, consumer surplus decreases compared to the asymmetric service scenario. In comparison to a scenario without consumer credit service, when the consumer credit service is exclusively available to consumers purchasing platform products, third-party sellers tend to reduce their product price to counteract the platform's competitive advantage, which ultimately benefits consumers. However, when the platform shares the service to include third-party sellers, consumers' willingness to pay increases across both channels, leading to higher retail prices in both channels. Consequently, consumer surplus is significantly diminished compared to the asymmetric service scenario. This finding reveals a counterintuitive outcome: the asymmetric consumer credit service proves to be the most advantageous for consumers, while credit service sharing can potentially be detrimental. Therefore, if the

platform opts to share the consumer credit service, it becomes even more essential to place a heightened emphasis on enhancing the overall consumer experience. Some effective methods should be adapted to improve customer experience, such as recognizing consumers' needs, enhancing product quality, improving consumer service, etc. Additionally, governmental entities should establish regulatory measures to ensure fairness, transparency, and security in consumer credit service operations. These measures are essential for safeguarding consumer rights and promoting responsible lending practices.

·*How does the critical parameter of the consumer credit service affect equilibrium results?* We find that under the *IN* and *II* modes, the impacts of the repayment rate on equilibrium results are different. In the *IN* mode, characterized by the strategic effect of asymmetric consumer credit services, retail prices for both products decrease as the repayment rate increases, indicating that a higher repayment rate can benefit consumers. However, in the *II* mode, with symmetric consumer credit services, retail prices for both products increase as the repayment rate rises, suggesting that a higher repayment rate is detrimental to consumers. Therefore, when a platform decides to share credit services with third-party sellers, we recommend implementing effective measures to reduce default risk. These measures could include enhancing the accuracy of consumer credit ratings, offering flexible repayment options, and rewarding timely repayments.

Overall, these findings provide valuable insights and guidance for the operations of hybrid e-commerce platforms, particularly regarding the provision and sharing of consumer credit services. Despite the intriguing results presented in this paper, certain limitations should be acknowledged. First, this study considers only one third-party seller who competes and cooperates with the platform. Future research could explore the competitive scenarios among multiple sellers in the context of consumer credit services. Additionally, this study assumes that consumer credit services are provided solely by the hybrid platform. However, in practice, we observe the rise of financial technology start-ups such as Affirm, Afterpay, and Klarna. These companies collaborate with retailers to offer installment services at checkout and charge a commission fee to retailers. Cooperation with financial technology companies, including consumer information sharing, may be an interesting research direction. Despite these limitations, this research advances our understanding of the impacts of consumer credit services on platforms and third-party sellers and contributes to the emerging intersection of financial innovation and platform operations.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

Endnotes

[1] See https://www.researchandmarkets.com/report/china-buy-now-pay-later-market.

[2] See https://www.cnbc.com/2020/12/14/buy-now-pay-later-plans-are-booming-in-the-covid-economy.html.

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[5] See https://www1.hkexnews.hk/listedco/listconews/sehk/2020/1026/2020102600165.pdf.

[6] See https://www.marketplacepulse.com/stats/amazon-third-party-seller-services-sales.

[7] See https://www.statista.com/statistics/259782/third-party-seller-share-of-amazon-platform/.

[8] See https://kr-asia.com/how-ai-and-vast-data-support-ant-groups-financial-empire.

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Appendix A

A.1 | Proof of Lemma 1

Considering the price w_a and demand function $D_a^{NN} = 1 - \frac{p_a^{NN} - p_b^{NN}}{1-\lambda}$, $D_b^{NN} = \frac{p_a^{NN} - p_b^{NN}}{1-\lambda} - \frac{p_b^{NN}}{\lambda}$, we adopt the backward induction and first solve the problem of the platform and the third-party seller:

$$\max_{p_a^{NN}} \pi_P^{NN} = \left(p_a^{NN} - w_a^{NN}\right) D_a^{NN} + \alpha p_b^{NN} D_b^{NN}$$

$$\max_{p_b^{NN}} \pi_T^{NN} = (1-\alpha) p_b^{NN} D_b^{NN}$$

Calculating the first and second derivatives of p_a^{NN} and p_b^{NN}, we have $\frac{\partial \pi_P^{NN}}{\partial p_a^{NN}} = \frac{-1+\lambda+2p_a^{NN}-(1+\alpha)p_b^{NN}-w_a^{NN}}{-1+\lambda}$, $\frac{\partial \pi_T^{NN}}{\partial p_b^{NN}} = \frac{(1-\alpha)\left(\lambda p_a^{NN}-2p_b^{NN}\right)}{(1-\lambda)\lambda}$, $\frac{\partial^2 \pi_P^{NN}}{\partial \left(p_a^{NN}\right)^2} = \frac{2}{-1+\lambda} < 0$, and $\frac{\partial^2 \pi_T^{NN}}{\partial \left(p_b^{NN}\right)^2} = -\frac{2(1-\alpha)}{(1-\lambda)\lambda} < 0$. By calculating the first derivatives to be 0, we have $p_a^{NN} = \frac{2\left(1-\lambda+w_a^{NN}\right)}{4-\lambda-\alpha\lambda}$. $p_b^{NN} = \frac{\lambda\left(1-\lambda+w_a^{NN}\right)}{4-\lambda-\alpha\lambda}$. Then we substitute p_a^{NN} and p_b^{NN} into the supplier A's problem:

$$\max_{w_a^{NN}} \pi_W^{NN} = w_a^{NN} D_a^{NN}$$

Calculating the first and second derivatives of w_a, we have $\frac{\partial \pi_W^{NN}}{\partial w_a^{NN}} = \frac{(1-\lambda)(2-\alpha\lambda)+2(-2+\lambda)w_a^{NN}}{(1-\lambda)(4-\lambda-\alpha\lambda)}$, and $\frac{\partial^2 \pi_W^{NN}}{\partial \left(w_a^{NN}\right)^2} = \frac{2(-2+\lambda)}{(1-\lambda)(4-\lambda-\alpha\lambda)} < 0$. Making $\frac{\partial \pi_W^{NN}}{\partial w_a^{NN}} = 0$, we have $w_a^{NN*} = \frac{(1-\lambda)(2-\alpha\lambda)}{2(2-\lambda)}$. Substituting the expression of w_a^{NN*} into p_a^{NN} and p_b^{NN}, we get $p_a^{NN*} = \frac{(1-\lambda)(6-(2+\alpha)\lambda)}{(2-\lambda)(4-\lambda-\alpha\lambda)}$ and $p_b^{NN*} = \frac{(1-\lambda)\lambda(6-(2+\alpha)\lambda)}{2(2-\lambda)(4-\lambda-\alpha\lambda)}$. It is easy to find that the equilibrium solutions meet the demand constraint $v_b < v_a$ (i.e., $\frac{p_b^{NN*}}{\lambda} < p_a^{NN*}$), when $0 < \lambda < 1$ and $0 < \alpha < 1/5$. Finally, we get the optimal profits and consumer surplus by calculating $CS^{NN*} = \int_{v_b}^{\frac{p_a^{NN*}-p_b^{NN*}}{1-\lambda}} \left(\lambda v - p_b^{NN*}\right) dv + \int_{\frac{p_a^{NN*}-p_b^{NN*}}{1-\lambda}}^{1} \left(v - p_a^{NN*}\right) dv$.

A.2 | Proof of Lemma 2

Similar to the NN mode, calculating the first and second derivatives of p_a^{IN} and p_b^{IN}, we have $\frac{\partial \pi_P^{IN}}{\partial p_a^{IN}} = \frac{\theta(1-\lambda)-2(1-\gamma)\theta p_a^{IN}+(\alpha(1-\gamma)+\theta)p_b^{IN}+(1-\gamma)w_a^{IN}}{1-\lambda}$, $\frac{\partial \pi_T^{IN}}{\partial p_b^{IN}} = \frac{(1-\alpha)\left((1-\gamma)\lambda p_a^{IN}-2p_b^{IN}\right)}{(1-\lambda)\lambda}$, $\frac{\partial^2 \pi_P^{IN}}{\partial \left(p_a^{IN}\right)^2} = -\frac{2(1-\gamma)\theta}{1-\lambda} < 0$, and $\frac{\partial^2 \pi_T^{IN}}{\partial \left(p_b^{IN}\right)^2} = -\frac{2(1-\alpha)}{(1-\lambda)\lambda} < 0$. By calculating the first derivatives to 0, we have: $p_a^{IN} = \frac{2\left(\theta(-1+\lambda)+(-1+\gamma)w_a^{IN}\right)}{(1-\gamma)(-4\theta+(\alpha-\alpha\gamma+\theta)\lambda)}$. $p_b^{IN} = \frac{\lambda\left(\theta(-1+\lambda)+(-1+\gamma)w_a^{IN}\right)}{-4\theta+(\alpha-\alpha\gamma+\theta)\lambda}$. Then we substitute p_a^{IN} and p_b^{IN} into π_W^{IN}.

Calculating the first and second derivatives of w_a, we have $\frac{\partial \pi_W^{IN}}{\partial w_a^{IN}} = -\frac{(\lambda-1)(2\theta+\alpha(-1+\gamma)\lambda)+2(1-\gamma)(2-\lambda)w_a^{IN}}{(1-\lambda)(4\theta-(\alpha-\alpha\gamma+\theta)\lambda)}$, and $\frac{\partial^2 \pi_W^{IN}}{\partial \left(w_a^{IN}\right)^2} = -\frac{2(1-\gamma)(2-\lambda)}{(1-\lambda)(4\theta-(\alpha-\alpha\gamma+\theta)\lambda)} < 0$, when $\frac{1}{2} < \theta < 1$. Making $\frac{\partial \pi_W^{IN}}{\partial w_a^{IN}} = 0$, we obtain $w_a^{IN*} = \frac{(1-\lambda)(2\theta+\alpha(\gamma-1)\lambda)}{2(1-\gamma)(2-\lambda)}$. Substituting the expression of w_a^{IN*} into p_a^{IN} and p_b^{IN}, we get $p_a^{IN*} = \frac{(1-\lambda)(2\theta(-3+\lambda)+\alpha(1-\gamma)\lambda)}{(1-\gamma)(2-\lambda)(-4\theta+(\alpha-\alpha\gamma+\theta)\lambda)}$, $p_b^{IN*} = \frac{(1-\lambda)\lambda(2\theta(-3+\lambda)+\alpha(1-\gamma)\lambda)}{2(2-\lambda)(-4\theta+(\alpha-\alpha\gamma+\theta)\lambda)}$. It is easy to find that the equilibrium results meet the demand constraint $v_b < v_a$ (i.e., $\frac{p_b^{IN*}}{\lambda} < (1-\gamma)p_a^{IN*}$). Finally, we get the optimal profits and consumer surplus by calculating $CS^{IN*} = \int_{\frac{p_b^{IN*}}{\lambda}}^{\frac{(1-\gamma)p_a^{IN*}-p_b^{IN*}}{1-\lambda}} \left(\lambda v - p_b\right) dv + \int_{\frac{(1-\gamma)p_a^{IN*}-p_b^{IN*}}{1-\lambda}}^{1} \left(v - p_a^{IN*} + \gamma p_a^{IN*}\right) dv$.

A.3 | Proof of Lemma 3

First, by taking derivatives, we can easily get $\frac{\partial p_a^{IN*}}{\partial \theta} = \frac{\alpha(-1+\lambda)\lambda}{(-4\theta+(\alpha-\alpha\gamma+\theta)\lambda)^2} < 0$, $\frac{\partial p_b^{IN*}}{\partial \theta} = \frac{\alpha(-1+\lambda)(1-\gamma)\lambda^2}{2(-4\theta+(\alpha-\alpha\gamma+\theta)\lambda)^2} < 0$, and $\frac{\partial w_a^{IN*}}{\partial \theta} = \frac{1-\lambda}{(1-\gamma)(2-\lambda)} > 0$. Next, taking the derivative of D_a^{IN*}, D_b^{IN*}, and CS^{IN*} gives $\frac{\partial D_a^{IN*}}{\partial \theta} = \frac{\alpha(1-\gamma)(2-\lambda)\lambda}{2(-4\theta+(\alpha-\alpha\gamma+\theta)\lambda)^2} > 0$, $\frac{\partial D_b^{IN*}}{\partial \theta} = \frac{\alpha(-1+\gamma)\lambda}{2(-4\theta+(\alpha-\alpha\gamma+\theta)\lambda)^2} < 0$, and $\frac{\partial CS^{IN*}}{\partial \theta} = \frac{\alpha(1-\gamma)(1-\lambda)\lambda(2\theta(4+(1-\lambda)\lambda)-\alpha(1-\gamma)(4-\lambda)\lambda)}{4(2-\lambda)(4\theta-(\alpha-\alpha\gamma+\theta)\lambda)^3}$. As $\theta > \frac{1}{2}$ and $0 < \alpha < \frac{1}{5}$, we can get $2\theta(4+(1-\lambda)\lambda)-\alpha(1-\gamma)(4-\lambda)\lambda > 0$ and $4\theta-(\alpha-\alpha\gamma+\theta)\lambda > 0$. Therefore, $\frac{\partial CS^{IN*}}{\partial \theta} > 0$.

Third, by taking derivatives of π_P^{IN*}, we have $\frac{\partial \pi_P^{IN*}}{\partial \theta} = \frac{(1-\lambda)H_1}{4(1-\gamma)(2-\lambda)(\theta(4-\lambda)-\alpha(1-\gamma)\lambda)^3}$, where $H_1 = 4\theta^3(4-\lambda)(2-\lambda) - 12\alpha(1-\gamma)\theta^2(2-\lambda)\lambda + \alpha^3 (1-\gamma)^3\lambda^4 + \alpha^2(1-\gamma)^2\theta\lambda^2(4-(6-\lambda)\lambda)$. Moreover, we can get that $\frac{\partial H_1}{\partial \theta} = 12\theta^2(4-\lambda)(2-\lambda) - 24\alpha(1-\gamma)\theta(2-\lambda)\lambda + \alpha^2(1-\gamma)^2\lambda^2(4-(6-\lambda)\lambda)$. As $\theta > \frac{1}{2} > \frac{24\alpha(1-\gamma)(2-\lambda)\lambda}{24(4-\lambda)(2-\lambda)} = \frac{\alpha(1-\gamma)\lambda}{(4-\lambda)}$, when $\theta = \frac{1}{2}$, $\frac{\partial H_1}{\partial \theta}$ gets the minimum. $\left.\frac{\partial H_1}{\partial \theta}\right|_{\theta=\frac{1}{2}} = 24+\lambda(3(-6+\lambda)+\alpha(-1+\gamma)(24+\lambda(-12+\alpha(-1+\gamma)(4+(-6+\lambda)\lambda)))) > 0$ when $0 < \lambda < 1$ and $0 < \alpha < \frac{1}{5}$. Therefore, $\frac{\partial H_1}{\partial \theta} > 0$, and when $\theta = \frac{1}{2}$, H_1 gets the minimum. $H_1\big|_{\theta=\frac{1}{2}} = \frac{1}{2}(8+\lambda(-6+\lambda+\alpha(-1+\gamma)(12-\lambda(6+\alpha(1-\gamma)(4-\lambda(6-\lambda-2\alpha\lambda+2\alpha\gamma\lambda))))))$. As $6+\alpha(1-\gamma)(4-\lambda(6-\lambda-2\alpha\lambda+2\alpha\gamma\lambda)) > 0$ when $0 < \alpha < \frac{1}{5}$, $\alpha(-1+\gamma)(12-\lambda(6+\alpha(1-\gamma)(4-\lambda(6-\lambda-2\alpha\lambda+2\alpha\gamma\lambda)))) > -12\alpha(1-\gamma) > -3$. Therefore, we can get $H_1\big|_{\theta=\frac{1}{2}} > \frac{1}{2}(8+\lambda(-6+\lambda-3)) = \frac{1}{2}(\lambda-1)(\lambda-8) > 0$ when $0 < \lambda < 1$. Thus, we can get $H_1 > 0$ and $\frac{\partial \pi_P^{IN*}}{\partial \theta} = \frac{(1-\lambda)H_1}{4(1-\gamma)(2-\lambda)(\theta(4-\lambda)-\alpha(1-\gamma)\lambda)^3} > 0$.

Finally, note that $\pi_W^{IN*} = w_a^{IN*} D_a^{IN*}$, and both w_a^{IN*} and D_a^{IN*} increase in θ. Hence, π_W^{IN*} increases in θ. $\pi_T^{IN*} = (1-\alpha)p_b^{IN*}D_b^{IN*}$, and both p_b^{IN*} and D_b^{IN*} decrease in θ. Therefore, π_T^{IN*} decreases in θ.

A.4 | Proof of Proposition 1

From Lemma 3(*iii*), we obtain that π_P^{IN*} increases in θ. By solving $\pi_P^{IN*} = \pi_P^{NN*}$, we can get the unique solution $\theta_0 = 1 - \gamma$. Therefore, there is $\pi_P^{IN*} > \pi_P^{NN*}$ if $\theta > 1 - \gamma$. Mathematically, it is easy to get that $\pi_P^{IN*} > \pi_P^{NN*}$ when $\gamma > 1 - \theta$.

A.5 | Proof of Proposition 2

Under the condition $\gamma > 1 - \theta$, by comparing equilibrium results under NN mode with that under IN mode, we have the following equilibrium results:

$p_a^{IN*} - p_a^{NN*} = \frac{1-\lambda}{2-\lambda}\left(\frac{2\theta(3-\lambda)-\alpha(1-\gamma)\lambda}{(1-\gamma)(\theta(4-\lambda)-\alpha(1-\gamma)\lambda)} - \frac{2(3-\lambda)-\alpha\lambda}{4-\lambda-\alpha\lambda}\right)$. From Lemma 3(*i*), we get that p_a^{IN*} decreases in θ, so $\frac{2\theta(3-\lambda)-\alpha(1-\gamma)\lambda}{(1-\gamma)(\theta(4-\lambda)-\alpha(1-\gamma)\lambda)} > \frac{2(3-\lambda)-\alpha(1-\gamma)\lambda}{(1-\gamma)((4-\lambda)-\alpha(1-\gamma)\lambda)}$. We can readily get that $\frac{2(3-\lambda)-\alpha(1-\gamma)\lambda}{(1-\gamma)((4-\lambda)-\alpha(1-\gamma)\lambda)} - \frac{2(3-\lambda)-\alpha\lambda}{4-\lambda-\alpha\lambda} = \frac{\gamma\left(24+2(-7+3\alpha(-2+\gamma))\lambda+(2+\alpha(4+\alpha-(2+\alpha)\gamma))\lambda^2\right)}{(1-\gamma)((4-\lambda)-\alpha(1-\gamma)\lambda)(4-\lambda-\alpha\lambda)} > 0$. Therefore, $p_a^{IN*} > p_a^{NN*}$.

$$w_a^{IN*} - w_a^{NN*} = \frac{(\gamma - 1 + \theta)(1-\lambda)}{(1-\gamma)(2-\lambda)} > 0.$$

$$p_b^{IN*} - p_b^{NN*} = -\frac{\alpha(\gamma - 1 + \theta)(1-\lambda)\lambda^2}{2(4-\lambda-\alpha\lambda)(\theta(4-\lambda)-\alpha(1-\gamma)\lambda)} < 0.$$

$$D_a^{IN*} - D_a^{NN*} = \frac{\alpha(\gamma - 1 + \theta)(2-\lambda)\lambda}{2(4-\lambda-\alpha\lambda)(\theta(4-\lambda)-\alpha(1-\gamma)\lambda)} > 0.$$

$$D_b^{IN*} - D_b^{NN*} = -\frac{\alpha(\gamma - 1 + \theta)\lambda}{2(4-\lambda-\alpha\lambda)(\theta(4-\lambda)-\alpha(1-\gamma)\lambda)} < 0.$$

Note that $\pi_W = w_a D_a$, $w_a^{IN*} > w_a^{NN*}$, and $D_a^{IN*} > D_a^{NN*}$. Hence, $\pi_W^{IN*} > \pi_W^{NN*}$. As $\pi_T = (1-\alpha)p_b D_b$, $p_b^{IN*} < p_b^{NN*}$, and $D_b^{IN*} < D_b^{NN*}$, $\pi_T^{IN*} < \pi_T^{NN*}$. From Lemma 3(*iv*), we get that CS^{IN*} increases in θ. By solving $CS^{IN*} = CS^{NN*}$, we can get the unique solution $\theta_0 = 1 - \gamma$, which means that when $\gamma > 1 - \theta, CS^{IN*} > CS^{NN*}$.

A.6 | Proof of Lemma 4

Calculating the first and second derivatives of p_a^{II} and p_b^{II}, we have $\frac{\partial \pi_P^{II}}{\partial p_a^{II}} = \frac{\theta(1-\lambda)+2(1-\gamma)\theta p_a^{II}-(1-\gamma)(\alpha+\beta+2\theta-1)p_b^{II}-(1-\gamma)w_a^{II}}{1-\lambda}$, $\frac{\partial \pi_T^{II}}{\partial p_b^{II}} = \frac{(1-\alpha-\beta)(1-\gamma)\left(\lambda p_a^{II}-2p_b^{II}\right)}{(1-\lambda)\lambda}$, $\frac{\partial^2 \pi_P^{II}}{\partial (p_a^{II})^2} = -\frac{2(1-\gamma)\theta}{1-\lambda} < 0$, and $\frac{\partial^2 \pi_T^{II}}{\partial (p_b^{II})^2} = -\frac{2(1-\alpha-\beta)(1-\gamma)}{(1-\lambda)\lambda} < 0$. By calculating the first derivatives to be 0, we have $p_a^{II} = \frac{2\left(\theta-\theta\lambda-(\gamma-1)w_a^{II}\right)}{(1-\gamma)(2\theta(2-\lambda)+(1-\alpha-\beta)\lambda)}$ and $p_b^{II} = \frac{\lambda\left(\theta-\theta\lambda-(\gamma-1)w_a^{II}\right)}{(1-\gamma)(2\theta(2-\lambda)+(1-\alpha-\beta)\lambda)}$. Then we substitute p_a^{II} and p_b^{II} into π_W^{II}.

Calculating the first and second derivatives of w_a^{II}, we have $\frac{\partial \pi_W^{II}}{\partial w_a^{II}} = \frac{(1-\lambda)(2\theta-(\alpha+\beta+\theta-1)\lambda)-2(1-\gamma)(2-\lambda)w_a^{II}}{(1-\lambda)(2\theta(2-\lambda)+(1-\alpha-\beta)\lambda)}$, and $\frac{\partial^2 \pi_W^{II}}{\partial \left(w_a^{II}\right)^2} = -\frac{2(1-\gamma)(2-\lambda)}{(1-\lambda)(2\theta(2-\lambda)+(1-\alpha-\beta)\lambda)} < 0$. Making $\frac{\partial \pi_W^{II}}{\partial w_a^{II}} = 0$, we get $w_a^{II*} = \frac{(1-\lambda)(2\theta-(M+\theta)\lambda)}{2(1-\gamma)(2-\lambda)}$, where $M = \alpha + \beta - 1$. Substituting the expression of w_a^{II*} into p_a^{II} and p_b^{II}, we get $p_a^{II*} = \frac{(1-\lambda)(3\theta(-2+\lambda)+M\lambda)}{(1-\gamma)(2-\lambda)(2\theta(-2+\lambda)+M\lambda)}$, $p_b^{II*} = \frac{(1-\lambda)\lambda(3\theta(-2+\lambda)+M\lambda)}{2(1-\gamma)(2-\lambda)(2\theta(-2+\lambda)+M\lambda)}$. It is easy to find that the solutions meet the demand constraint $v_b < v_a$ (i.e., $\frac{(1-\gamma)p_b^{II*}}{\lambda} < (1-\gamma)p_a^{II*}$). Finally, we get the optimal profits and consumer surplus by calculating $CS^{II*} = \int_{\frac{(1-\gamma)p_b^{II*}}{\lambda}}^{\frac{(1-\gamma)\left(p_a^{II*}-p_b^{II*}\right)}{1-\lambda}} \left(\lambda v - p_b^{II*} + \gamma p_b^{II*}\right)dv + \int_{\frac{(1-\gamma)\left(p_a^{II*}-p_b^{II*}\right)}{1-\lambda}}^{1} \left(v - p_a^{II*} + \gamma p_a^{II*}\right)dv$.

A.7 | Proof of Lemma 5

By taking the derivatives of p_a^{II*}, p_b^{II*}, and w_a^{II*}, we can easily get $\frac{\partial p_a^{II*}}{\partial \theta} = \frac{(1-\alpha-\beta)(1-\lambda)\lambda}{(1-\gamma)(2\theta(-2+\lambda)+(-1+\alpha+\beta)\lambda)^2} > 0$, $\frac{\partial p_b^{II*}}{\partial \theta} = \frac{(1-\alpha-\beta)(1-\lambda)\lambda^2}{2(1-\gamma)(2\theta(-2+\lambda)+(-1+\alpha+\beta)\lambda)^2} > 0$, and $\frac{\partial w_a^{II*}}{\partial \theta} = \frac{1-\lambda}{2(1-\gamma)} > 0$. Similarly, taking the derivative of D_a^{II*}, D_b^{II*}, and π_W^{II*} gives $\frac{\partial D_a^{II*}}{\partial \theta} = -\frac{(1-\alpha-\beta)(2-\lambda)\lambda}{2(2\theta(-2+\lambda)+(-1+\alpha+\beta)\lambda)^2} < 0$, $\frac{\partial D_b^{II*}}{\partial \theta} = \frac{(1-\alpha-\beta)\lambda}{2(2\theta(-2+\lambda)+(-1+\alpha+\beta)\lambda)^2} > 0$, $\frac{\partial \pi_W^{II*}}{\partial \theta} = \frac{\theta(2-\lambda)(1-\lambda)(2\theta+(1-\alpha-\beta-\theta)\lambda)}{2(1-\gamma)(2\theta(-2+\lambda)+(-1+\alpha+\beta)\lambda)^2} > 0$.

Note that in II mode $\pi_T^{II*} = (1-\alpha-\beta)p_b^{II*}D_b^{II*}$, both p_b^{II*} and D_b^{II*} increase in θ. Hence, π_T^{II*} increases in θ. By taking the derivative of π_P^{II*}, we find that $\frac{\partial \pi_P^{II*}}{\partial \theta} = -\frac{(1-\lambda)(2\theta+(M-\theta)\lambda)\left(-M^2\lambda^3+\theta(2-\lambda)(2\theta(2-\lambda)+M\lambda)(-4+\lambda(-11+2\lambda))\right)}{4(1-\gamma)(\lambda-2)^2(2\theta(2-\lambda)+M\lambda)^3}$, where $M = 1 - \alpha - \beta$. When $\frac{1}{2} < \theta < 1,\ \ 0 < \alpha < \frac{1}{5}$ and $0 < \beta < \frac{1}{10}$, $(2\theta(2-\lambda)+M\lambda)(-4+\lambda(-11+2\lambda)) < 0$ and $2\theta(2-\lambda)+M\lambda > 0$. Therefore, $\frac{\partial \pi_P^{II*}}{\partial \theta} > 0$.

Finally, we have $\frac{\partial CS^{II*}}{\partial \theta} = \frac{M(1-\lambda)\lambda(-M(4-\lambda)\lambda+\theta(-2+\lambda)(4+\lambda))}{4(2-\lambda)(2\theta(2-\lambda)+M\lambda)^3} < 0$, where $M = 1 - \alpha - \beta$.

A.8 | Proof of Proposition 3

Similar to Lemma 5, we can easily get that $\frac{\partial p_b^{II*}}{\partial \beta} = \frac{\theta(1-\lambda)\lambda^2}{2(1-\gamma)(2\theta(-2+\lambda)+(-1+\alpha+\beta)\lambda)^2} > 0$ and $\frac{\partial D_b^{II*}}{\partial \beta} = \frac{2\theta\lambda}{(4\theta(-2+\lambda)-2(1-\alpha-\beta)\lambda)^2} > 0$.

$\frac{\partial \pi_T^{II*}}{\partial \beta} = \frac{(-1+\lambda)\lambda(3\theta(-2+\lambda)+(-1+\alpha+\beta)\lambda)\left(6\theta^2(2-\lambda)^2+3(1-\alpha-\beta)\theta(2-\lambda)\lambda+(1-\alpha-\beta)^2\lambda^2\right)}{4(1-\gamma)(2-\lambda)^2(2\theta(-2+\lambda)+(-1+\alpha+\beta)\lambda)^3}$. Since $6\theta^2(2-\lambda)^2+3(1-\alpha-\beta)\theta(2-\lambda)\lambda+(1-\alpha-\beta)^2\lambda^2>0$ and $2\theta(-2+\lambda)+(-1+\alpha+\beta)\lambda<0$, $\frac{\partial \pi_T^{II*}}{\partial \beta}<0$. Therefore, π_T^{II*} decreases in β.

As $\pi_P^{II*} = \left(\theta p_a^{II*} - w_a^{II*}\right)D_a^{II*} + (\alpha+\beta)p_b^{II*}D_b^{II*} - (1-\theta)p_b^{II*}D_b^{II*}$, we define $H_2 = \left(\theta p_a^{II*} - w_a^{II*}\right)D_a^{II*}$. We can get $\frac{\partial H_2}{\partial \beta} = \frac{(1-\lambda)\lambda\left(-11M\theta^2(2-\lambda)^2\lambda+6M^2\theta(-2+\lambda)\lambda^2-M^3\lambda^3+2\theta^3(2-\lambda)^2(-4+3\lambda)\right)}{4(1-\gamma)(2-\lambda)(2\theta(-2+\lambda)-M\lambda)^3}$, where $M=1-\alpha-\beta$. When $\frac{1}{2}<\theta<1, 0<\alpha<\frac{1}{5}$ and $0<\beta<\frac{1}{10}$, $11M\theta^2(2-\lambda)^2\lambda>0$, $6M^2\theta(-2+\lambda)\lambda^2<0$, $M^3\lambda^3>0$, and $2\theta^3(2-\lambda)^2(-4+3\lambda)<0$. Therefore, $\frac{\partial H_2}{\partial \beta}>0$. Note that under II mode, $\pi_P^{II*} = H_2+(\alpha+\beta)p_b^{II*}D_b^{II*}-(1-\theta)p_b^{II*}D_b^{II*} = H_2-\pi_T^{II*}+\theta p_b^{II*}D_b^{II*}$. H_2, p_b^{II*}, and D_b^{II*} increase in β; π_T^{II*} decreases in β; hence, π_P^{II*} increases in β.

Since π_P^{II*} increases in β, we can get a unique $\beta_1 = 1-\alpha\gamma-\theta$ by solving $\pi_P^{II*}=\pi_P^{IN*}$. Mathematically, there is $\pi_P^{II*}>\pi_P^{IN*}$ if $\beta>\beta_1$. This is equivalent to $\pi_P^{II*}>\pi_P^{IN*}$ if $\gamma>\frac{1-\theta-\beta}{\alpha}$. Note that the condition that the platform will launch the installment service is $\gamma>1-\theta$, we need to discuss three situations:

1. When $\beta \geq (1-\alpha)(1-\theta)$, we can have $\frac{1-\theta-\beta}{\alpha}\leq 1-\theta$. Therefore, under the condition $\gamma>1-\theta$, $\pi_P^{II*}>\pi_P^{IN*}$. Next, we need to get the condition under which the third-party seller is willing to use the service (i.e., $\pi_T^{II*}\geq\pi_T^{IN*}$). We define $g=\pi_T^{II*}-\pi_T^{IN*}=\frac{(1-\lambda)\lambda}{4(2-\lambda)^2}H_3$, where $H_3=\frac{(1-\alpha-\beta)(3\theta(-2+\lambda)-M\lambda)^2}{(1-\gamma)(2\theta(-2+\lambda)-M\lambda)^2}+\frac{(-1+\alpha)(-2\theta(-3+\lambda)+\alpha(-1+\gamma)\lambda)^2}{(-4\theta+(\alpha-\alpha\gamma+\theta)\lambda)^2}$, and $\frac{\partial H_3}{\partial \gamma}=\frac{M(3\theta(-2+\lambda)-M\lambda)^2}{(1-\gamma)^2(2\theta(-2+\lambda)-M\lambda)^2}+\frac{2(-1+\alpha)\alpha\lambda(-2\theta(-3+\lambda)+\alpha(-1+\gamma)\lambda)^2}{(-4\theta+(\alpha-\alpha\gamma+\theta)\lambda)^3}+\frac{2(-1+\alpha)\alpha\lambda(-2\theta(-3+\lambda)+\alpha(-1+\gamma)\lambda)}{(-4\theta+(\alpha-\alpha\gamma+\theta)\lambda)^2}$. Furthermore, $\frac{\partial^2 H_3}{\partial\gamma\partial\beta}=-\frac{(3\theta(-2+\lambda)+(-1+\alpha+\beta)\lambda)\left(6\theta^2(-2+\lambda)^2+3(-1+\alpha+\beta)\theta(-2+\lambda)\lambda+(-1+\alpha+\beta)^2\lambda^2\right)}{(1-\gamma)^2(2\theta(-2+\lambda)+(-1+\alpha+\beta)\lambda)^3}$. It is easy to check that $\frac{\partial^2 H_3}{\partial\gamma\partial\beta}<0$ and $\left.\frac{\partial H_3}{\partial\gamma}\right|_{\beta=\frac{1}{10}}>0$. Therefore, we have $\frac{\partial H_3}{\partial\gamma}>0$, $H_3\big|_{\gamma=1-\theta}=\frac{(-1+\alpha)(-6+(2+\alpha)\lambda)^2}{(-4+\lambda+\alpha\lambda)^2}-\frac{(-1+\alpha+\beta)(3\theta(-2+\lambda)+(-1+\alpha+\beta)\lambda)^2}{\theta(2\theta(-2+\lambda)+(-1+\alpha+\beta)\lambda)^2}$. That is, H_3 increases as γ increases. Since π_T^{II} decreases in β, H_3 decreases in β. Hence, when $\beta=(1-\alpha)(1-\theta)$, $H_3\big|_{\gamma=1-\theta}$ gets the maximum. We can get that $H_3\big|_{\gamma=1-\theta,\ \ \beta=(1-\alpha)(1-\theta)}=0$. Therefore, in this situation, when $\gamma\to 1-\theta$, there is $H_3\leq 0$. When $\gamma\to 1$, there is $H_3\to+\infty$. Since $\frac{\partial H_3}{\partial\gamma}>0$, there must exist γ_0 to make $H_3=0$ where γ_0 is the solution of $\pi_S^{II*}=\pi_S^{IN*}$.
2. When $1-\alpha-\theta<\beta<(1-\alpha)(1-\theta)$, $1-\theta<\frac{1-\theta-\beta}{\alpha}<1$. For the platform, it can be seen from the previous proof that $\pi_P^{II*}>\pi_P^{IN*}$ if $\gamma>\frac{1-\theta-\beta}{\alpha}$. For the third-party seller, $H_3\big|_{\gamma=1-\theta}>0$ when $\beta<(1-\alpha)(1-\theta)$. Since $\frac{\partial H_3}{\partial\gamma}>0$, we can get $H_3>0$, $\pi_S^{II*}>\pi_S^{IN*}$.
3. When $\beta\leq 1-\alpha-\theta$, $\frac{1-\theta-\beta}{\alpha}\geq 1$. In this situation, $\pi_P^{II*}\leq\pi_P^{IN*}$.

A.9 | Proof of Proposition 4

Since $\frac{\partial p_a^{II*}}{\partial\beta}=\frac{\theta(1-\lambda)\lambda}{(1-\gamma)(2\theta(-2+\lambda)+(-1+\alpha+\beta)\lambda)^2}>0$, that is, p_a^{II*} increases in β, we can get a unique solution $\beta_1=1-\alpha\gamma-\theta$ by solving $p_a^{II*}=p_a^{IN*}$, and $p_a^{II*}>p_a^{IN*}$ if $\beta>\beta_1$. Mathematically, this is equivalent to $p_a^{II*}>p_a^{IN*}$ if $\gamma>\frac{1-\theta-\beta}{\alpha}$. According to Proposition 3, only when $\gamma>\frac{1-\theta-\beta}{\alpha}$, the platform will be willing to share the installment service. Therefore, under the condition $\gamma>\frac{1-\theta-\beta}{\alpha}$, $p_a^{II*}>p_a^{IN*}$. Other proofs are similar to this process, we omit the proof process here.

The proof of p_b is special. As $\frac{\partial p_b^{II*}}{\partial\beta}=\frac{\theta(1-\lambda)\lambda^2}{2(1-\gamma)(2\theta(-2+\lambda)+(-1+\alpha+\beta)\lambda)^2}>0$, p_b^{II*} increases in β. According to Proposition 3, only when $\beta>1-\alpha\gamma-\theta$, the platform will be willing to share the installment service. We substitute $\beta=1-\alpha\gamma-\theta$ into p_b^{II*} and get $p_b^{II*}\Big|_{\beta=1-\alpha\gamma-\theta}=\frac{(1-\lambda)\lambda(2\theta(-3+\lambda)+\alpha(-1+\gamma)\lambda)}{2(1-\gamma)(2-\lambda)(-4\theta+(\alpha-\alpha\gamma+\theta)\lambda)}$. As $p_b^{IN*}=\frac{(1-\lambda)\lambda(2\theta(-3+\lambda)+\alpha(1-\gamma)\lambda)}{2(2-\lambda)(-4\theta+(\alpha-\alpha\gamma+\theta)\lambda)}$ and $0<\gamma<1$, we can readily get that $p_b^{II*}\Big|_{\beta=1-\alpha\gamma-\theta}>p_b^{IN*}$. Therefore, $p_b^{II*}>p_b^{IN*}$.

A.10 | Proof of Extension

The following Lemmas summarize the equilibrium results under *INC* and *IIC* modes.

Lemma A1. Under the *INC* mode, the equilibrium outcomes are as follows:

i. Wholesale price: $w_a^{INC*}=\frac{(1-\lambda)\left(2(1-\delta)(1+\gamma\delta)+2(1-\gamma(2-\delta))\delta\theta-\alpha(1-\gamma\delta)^2\lambda\right)}{2(1-\gamma\delta)^2(2-\lambda)}$.
ii. Retail prices: $p_a^{INC*}=\frac{(1-\lambda)\left(2(1-\delta(1-\theta))+\frac{2(1-\delta)(1+\gamma\delta)+2(1-\gamma(2-\delta))\delta\theta-\alpha(1-\gamma\delta)^2\lambda}{(1-\gamma\delta)(2-\lambda)}\right)}{P}$; $\mathrm{p_b^{INC*}}=\frac{(1-\lambda)\lambda(6-2\delta(3+\gamma-\gamma\delta+(-3+\gamma(2+\delta))\theta)-(1-\gamma\delta)(2+\alpha-\delta(2+\alpha\gamma-2\theta))\lambda)}{2(2-\lambda)\mathrm{P}}$.
iii. Demand: $D_a^{INC*}=\frac{2+2\delta(-1+\gamma(1-\delta(1-\theta)-2\theta)+\theta)-\alpha(1-\gamma\delta)^2\lambda}{2P}$; $D_b^{INC*}=\frac{6-2\delta(3+\gamma-\gamma\delta+(\gamma(2+\delta)-3)\theta)-(1-\gamma\delta)(2+\alpha-\delta(2+\alpha\gamma-2\theta))\lambda}{2(2-\lambda)P}$.
iv. Profits: $\pi_W^{INC*}=\frac{(1-\lambda)\left(2(1-\delta)(1+\gamma\delta)+2(1-\gamma(2-\delta))\delta\theta-\alpha(1-\gamma\delta)^2\lambda\right)^2}{4(1-\gamma\delta)^2(2-\lambda)P}$; $\pi_T^{INC*}=\frac{(1-\alpha)(1-\lambda)\lambda(6-2\delta(3+\gamma-\gamma\delta+(\gamma(2+\delta)-3)\theta)-(1-\gamma\delta)(2+\alpha-\delta(2+\alpha\gamma-2\theta))\lambda)^2}{4(2-\lambda)^2P^2}$; $\pi_P^{INC*}=\left(p_a^{INC*}-w_a^{INC*}\right)D_{a2}^{INC*}+\left(\theta p_a^{INC*}-w_a^{INC*}\right)D_{a1}^{INC*}+\alpha p_b^{INC*}D_b^{INC*}$,

where $D_{a1}^{INC*}=\left(1-\frac{(1-\gamma)p_a^{INC*}-p_b^{INC*}}{1-\lambda}\right)\delta$, $D_{a2}^{INC*}=\left(1-\frac{p_a^{INC*}-p_b^{INC*}}{1-\lambda}\right)(1-\delta)$ and $P=4(1-\delta+(1-\gamma)\delta\theta)-(1-\gamma\delta)(1+\alpha-\delta(1+\alpha\gamma-\theta))\lambda$.

The proof of Lemma A1 is similar to Lemma 2, so we omit the details.

Lemma A2. Under the *IIC* mode, the equilibrium outcomes are as follows:

i. Wholesale price:

$$w_a^{IIC*} = \frac{(1-\lambda)(2-\alpha\lambda-\delta(2-2\theta+\gamma\delta(1-\theta)(2-\lambda)-(1-\beta-\theta)\lambda-\gamma(2-2\theta(2-\lambda)-(2-\alpha-\beta)\lambda)))}{2(1-\gamma\delta)^2(2-\lambda)}.$$

ii. Retail prices:

$$p_a^{IIC*} = \frac{(1-\lambda)(6-(2+\alpha)\lambda+\delta(-2(3+\gamma-3\theta+2\gamma\theta)+\gamma\delta(1-\theta)(2-\lambda)+(3-\beta(1-\gamma)+\alpha\gamma-3\theta+2\gamma\theta)\lambda))}{(1-\gamma\delta)(2-\lambda)Q};$$

$$p_b^{IIC*} = \frac{(1-\lambda)\lambda(6-(2+\alpha)\lambda+\delta(-2(3+\gamma-3\theta+2\gamma\theta)+\gamma\delta(1-\theta)(2-\lambda)+(3-\beta(1-\gamma)+\alpha\gamma-3\theta+2\gamma\theta)\lambda))}{2(1-\gamma\delta)(2-\lambda)Q}.$$

iii. Demand:

$$D_{a1}^{IIC*} = \frac{(1-\delta)\left(-2+\alpha\lambda+\delta^2\gamma(-6+(3-4\gamma)\theta(2-\lambda)+(3-2\beta-2(1-\alpha-\beta)\gamma)\lambda)+\delta(2-2\theta-(1-\beta-\theta)\lambda+\gamma(6+2\theta(2-\lambda)-3\alpha\lambda-\beta\lambda))\right)}{2(-1+\gamma\delta)Q};$$

$$D_{a2}^{IIC*} = \frac{\delta(2(1-\delta)(1+\gamma(3-(3+\gamma)\delta))+2(1-\gamma)\delta(1+\gamma(2-3\delta))\theta-\lambda N)}{2(1-\gamma\delta)Q};$$

$$D_{b1}^{IIC*} = \frac{1}{2}\left(\frac{1-\delta}{(1-\gamma\delta)(2-\lambda)} + \frac{(1-\delta)(1-\delta(1-\theta))}{Q}\right);$$

$$D_{b2}^{IIC*} = \frac{1}{2}(1-\gamma)\delta\left(\frac{1}{(1-\gamma\delta)(2-\lambda)} + \frac{1-\delta+\delta\theta}{Q}\right).$$

iv. Profits:

$$\pi_W^{IIC*} = \frac{(1-\lambda)(2-\alpha\lambda-\delta(2-2\theta+\gamma\delta(1-\theta)(2-\lambda)-(1-\beta-\theta)\lambda-\gamma(2-2\theta(2-\lambda)-(2-\alpha-\beta)\lambda)))^2}{4(1-\gamma\delta)^2(2-\lambda)Q};$$

$$\pi_T^{IIC*} = \frac{(1-\alpha-\beta\delta+(\alpha+\beta-1)\gamma\delta)(1-\lambda)\lambda\left(6-2\delta(3+\gamma-3\theta+2\gamma\theta)+\delta^2\gamma(1-\theta)(2-\lambda)-(2+\alpha)\lambda+\delta(3-\beta(1-\gamma)+\alpha\gamma-3\theta+2\gamma\theta)\lambda\right)^2}{4(1-\gamma\delta)^2(2-\lambda)^2Q^2};$$

$\pi_P^{IIC*} = \left(p_a^{IIC*} - w_a^{IIC*}\right)D_{a1}^{IIC*} + \alpha p_b^{IIC*}D_{b1}^{IIC*} + \left(\theta p_a^{IIC*} - w_a^{IIC*}\right)D_{a2}^{IIC*} + (\alpha+\beta-(1-\theta))p_b^{IIC*}D_{b2}^{IIC*}$. where $N=\alpha+(2+\alpha)\gamma+\gamma\delta^2(3-2\beta(1-\gamma)-3\theta-\gamma(1-2\alpha-3\theta))+\delta\left(-1+\beta-\beta\gamma^2+\theta-\gamma(3+\alpha(3+\gamma)-\theta+2\gamma\theta)\right)$; $Q=4-(1+\alpha)\lambda+\delta(-4+2(1-\gamma)\theta(2-\lambda)+(2-\beta-(1-\alpha-\beta)\gamma)\lambda)$.

The proof of Lemma A1 is similar to Lemma 4, so we omit the details.